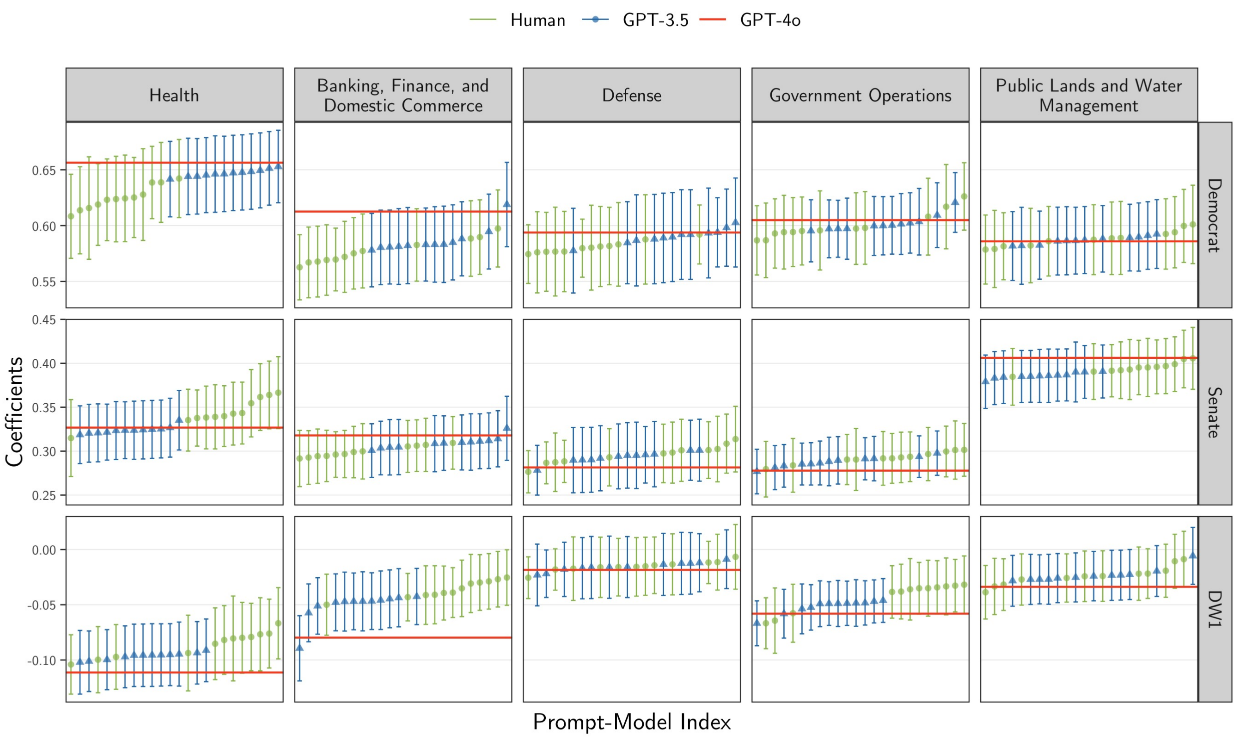
Results for Regressions with RHS Proxy

Sep 24, 2024

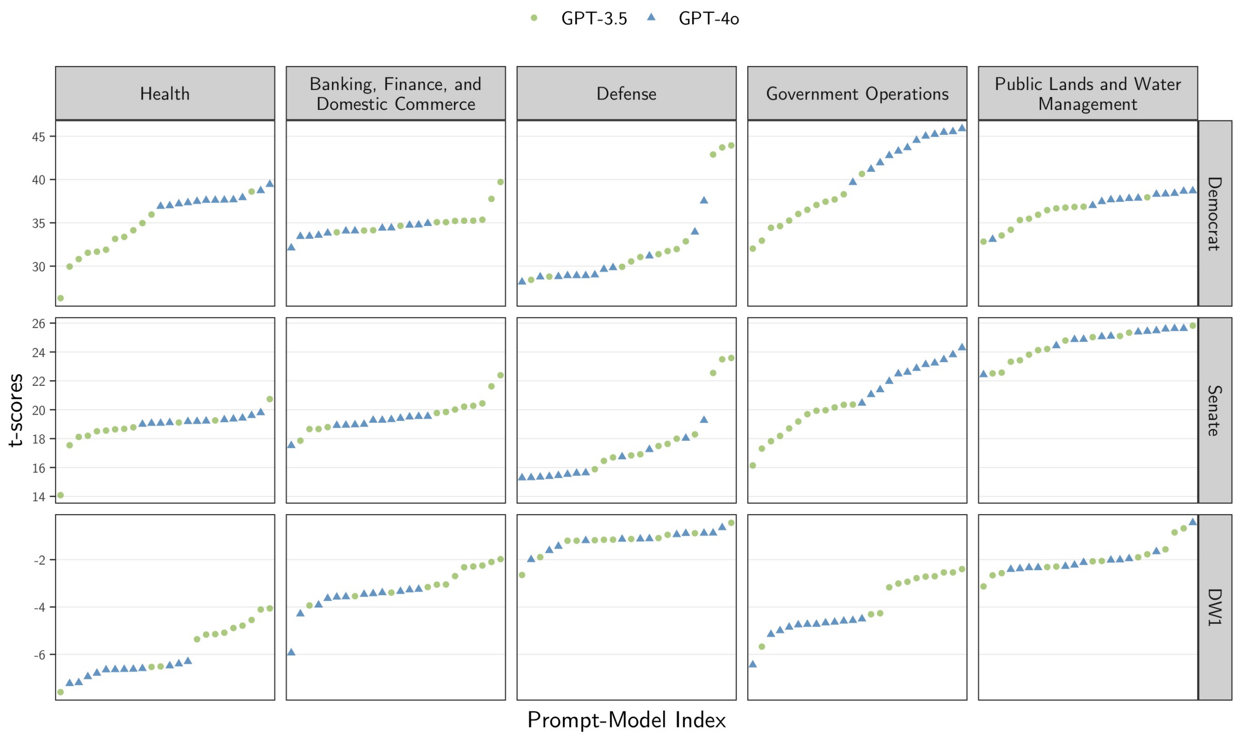
Proxy on the RHS

Figure 1: β_t vs. Prompt for all t. Proxy on the RHS.



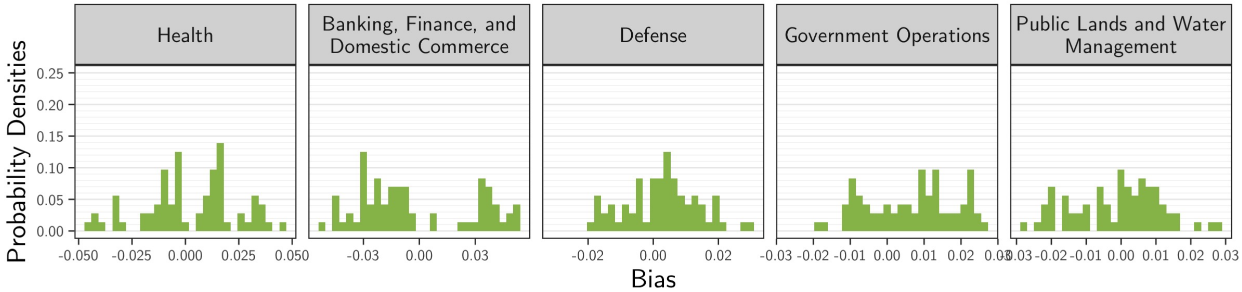
Note: The coefficients were obtained by fitting the model $V = Y^\top \beta = \sum_t Y_t \beta_t$ using 10,000 bills, with 95% confidence intervals computed via robust standard errors. Here, $V \in \{\text{Democrat; Senate; DW1}\}$ and $t \in \{$Health; Banking, Finance, and Domestic Commerce; Defense; Government Operations; Public Lands and Water Management; Other$\}$. The human-based coefficients $\beta^\star$ (red) were estimated using $Y_t^{\text{Human}} := 1\{Y^{\text{Human}} = t\}$, while LLM-based coefficients β^{LLM} (GPT-3.5 in green circles, GPT-4o in blue triangles) used $Y_{t,m,p}^{\text{LLM}} := 1\{Y_{m,p}^{\text{LLM}} = t\}$, with $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denoting the LLM model and $p \in \{1, \ldots, 12\}$ representing the prompt. In each subplot, the bias estimates were sorted in ascending order for clarity.

Figure 2: t-scores of each β_t vs. Prompt for all t. Proxy on the RHS.



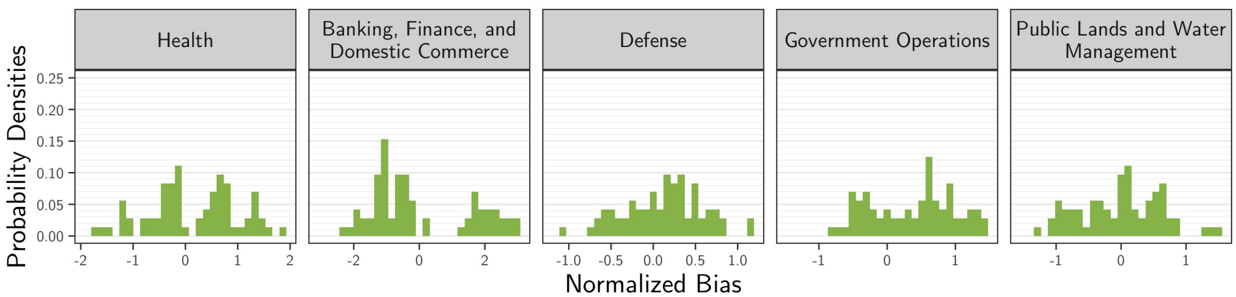
Note: The t-scores were obtained by fitting the model $V = Y^\top \beta = \sum_t Y_t \beta_t$ using 10,000 bills, with 95% confidence intervals computed via robust standard errors. Here, $V \in \{\text{Democrat}; \text{Senate}; \text{DW1}\}$ and $t \in \{$Health; Banking, Finance, and Domestic Commerce; Defense; Government Operations; Public Lands and Water Management; Other$\}$. The human-based coefficients $\beta^\star$ (red) were estimated using $Y_t^{\text{Human}} := 1\{Y^{\text{Human}} = t\}$, while LLM-based coefficients β^{LLM} (GPT-3.5 in green circles, GPT-4o in blue triangles) used $Y_{t,m,p}^{\text{LLM}} := 1\{Y_{m,p}^{\text{LLM}} = t\}$, with $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denoting the LLM model and $p \in \{1, \dots, 12\}$ representing the prompt. In each subplot, the bias estimates were sorted in ascending order for clarity.

Figure 3: Bias Density of β_t^{LLM} for all t. Proxy on the RHS.



Note: The bias estimates were calculated as the difference between the LLM-based coefficient β_t^{LLM} and the human-based coefficient $\beta_t^\star$, using a regression of the form $V = Y^\top \beta = \sum_t Y_t \beta_t$, where $V \in \{\text{Democrat}; \text{Senate}; \text{DW1}\}$ and $t \in \{$Health; Banking, Finance, and Domestic Commerce; Defense; Government Operations; Public Lands and Water Management; Other$\}$. For human-based coefficients, the covariates are $Y_t^{\text{Human}} := 1\{Y^{\text{Human}} = t\}$, while for LLM-based coefficients, the covariates are $Y_{t,m,p}^{\text{LLM}} := 1\{Y_{m,p}^{\text{LLM}} = t\}$, with $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denoting the LLM model and $p \in \{1, \ldots, 12\}$ representing the prompt. The human-based coefficient $\beta_t^\star$ was computed using the full 10,000 bills, whereas the LLM-based coefficient β_t^{LLM} was calculated as the average of $N = 1,000$ simulation runs, with each run's estimate based on a sample of 5,000 bills. The bias estimates are independent of the validation proportion η, though they were generated in the experiment run with $\eta = 5\%$.

Figure 4: Normalized Bias Density of β_t^{LLM} for all t. Proxy on the RHS.



Note: The normalized bias estimates were calculated as the difference between the LLM-based coefficient β_t^{LLM} and the human-based coefficient $\beta_t^\star$, divided by the coefficient's sample standard deviation computed across $N = 1000$ samples. All other variables are as defined in the previous figure. The bias estimates are independent of the validation proportion η, though they were generated in the experiment run with $\eta = 5\%$.

Figure 5: Bias Density of β_t, for t =Health, by Model and Proportion of Validation Samples. Proxy on the RHS.

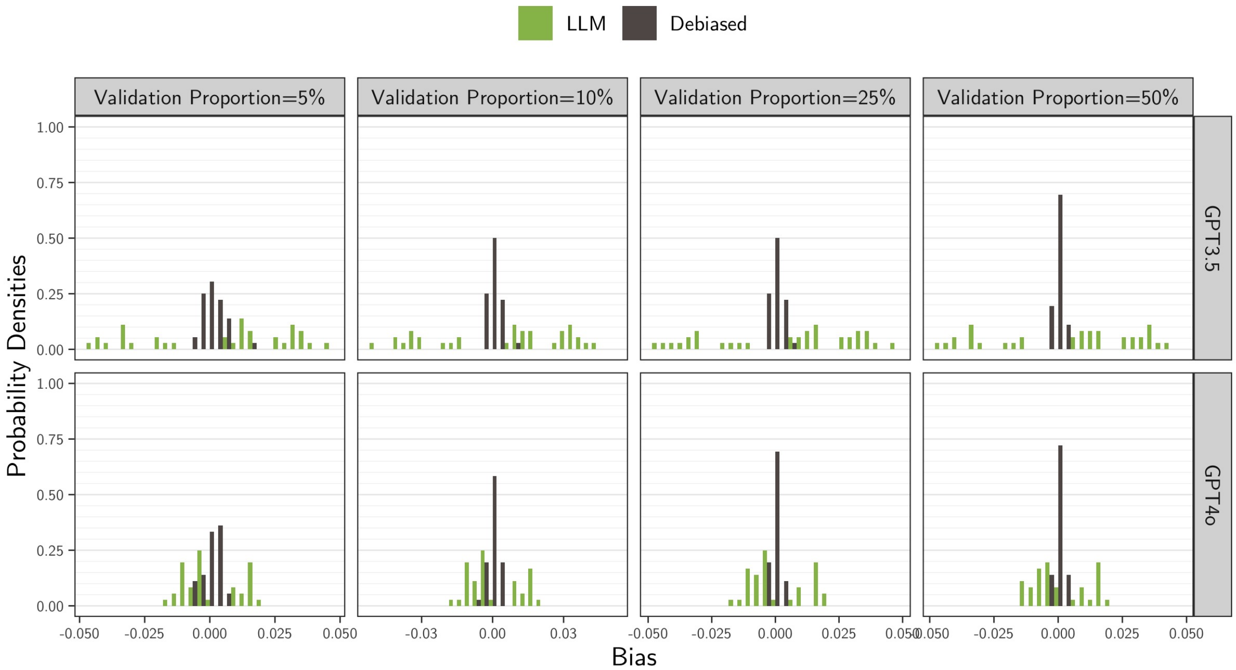


Figure 6: Bias Density of β_t, t =Banking, Finance, and Domestic Commerce, by Model and Proportion of Validation Samples. Proxy on the RHS.

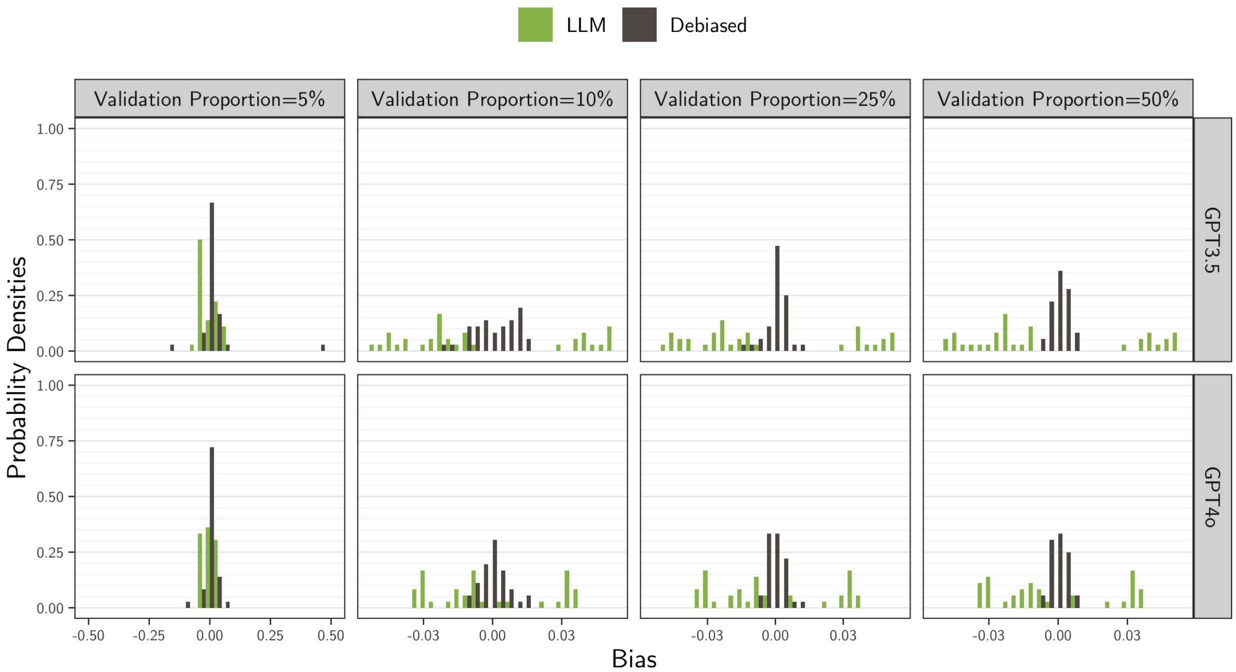


Figure 7: Bias Density of β_t, t =Defense, by Model and Proportion of Validation Samples. Proxy on the RHS.

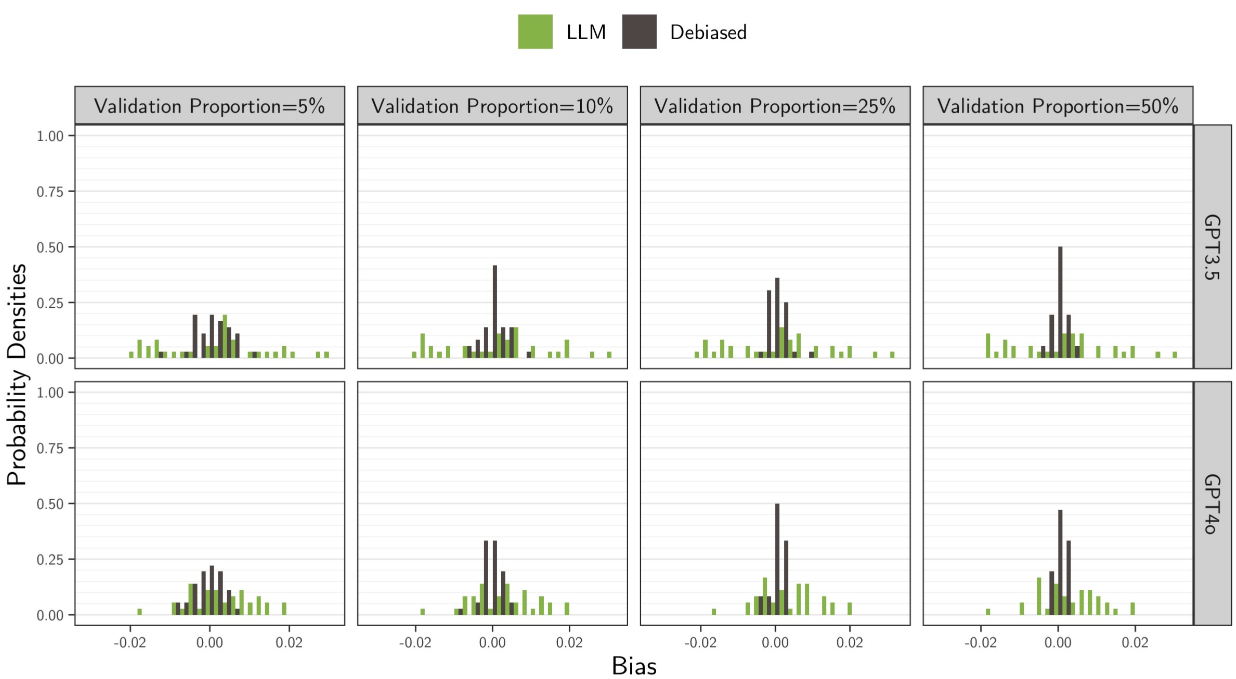


Figure 8: Bias Density of β_t, t =Government Operations, by Model and Proportion of Validation Samples. Proxy on the RHS.

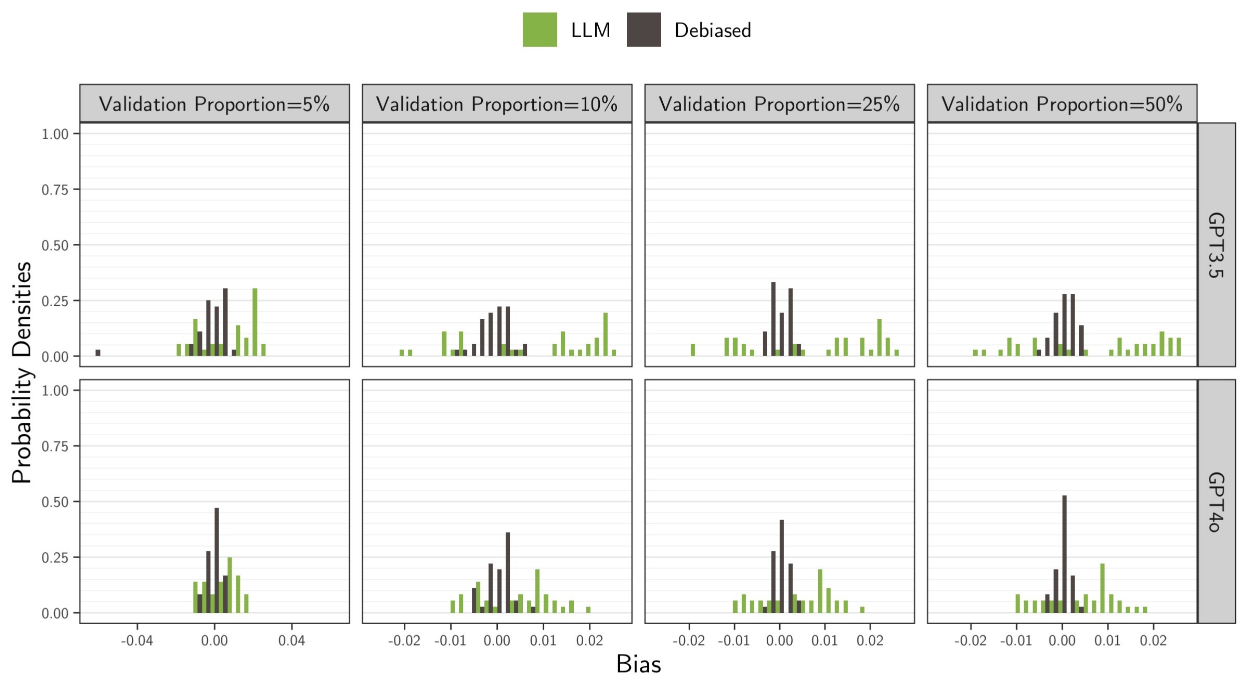


Figure 9: Bias Density of β_t, t =Public Lands and Water Management, by Model and Proportion of Validation Samples. Proxy on the RHS.

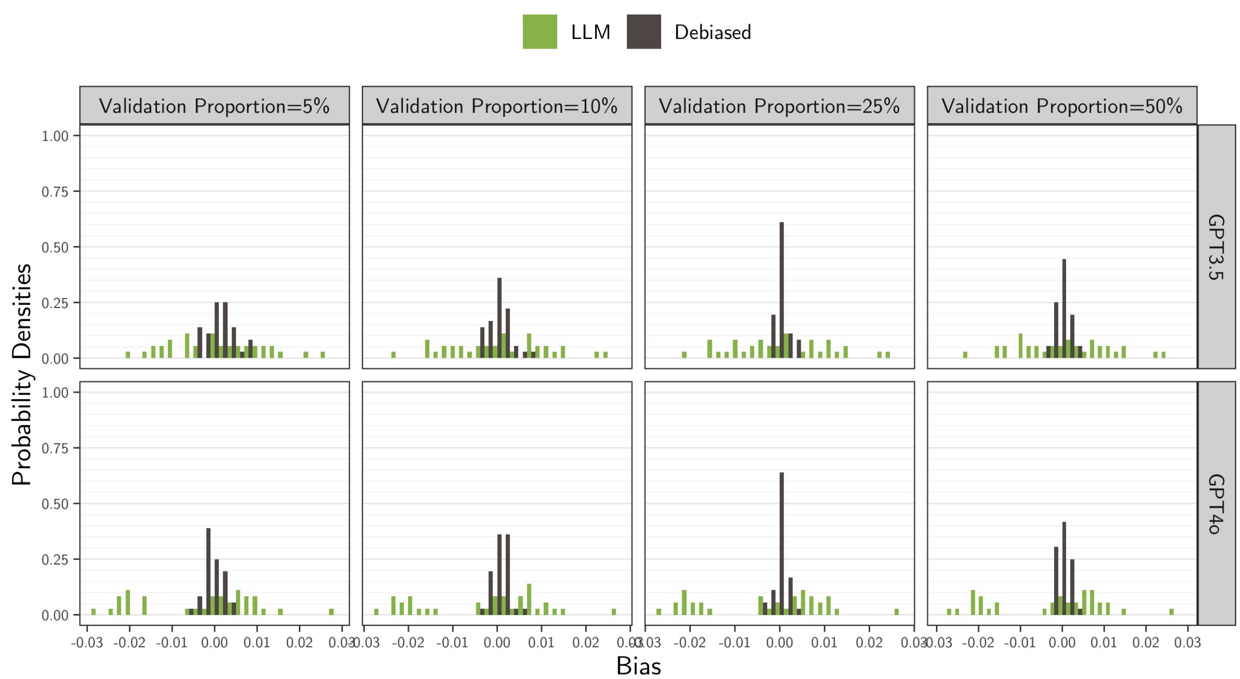


Figure 10: Normalized Bias Density of β_t, $t =$Health, by Model and Proportion of Validation Samples. Proxy on the RHS.

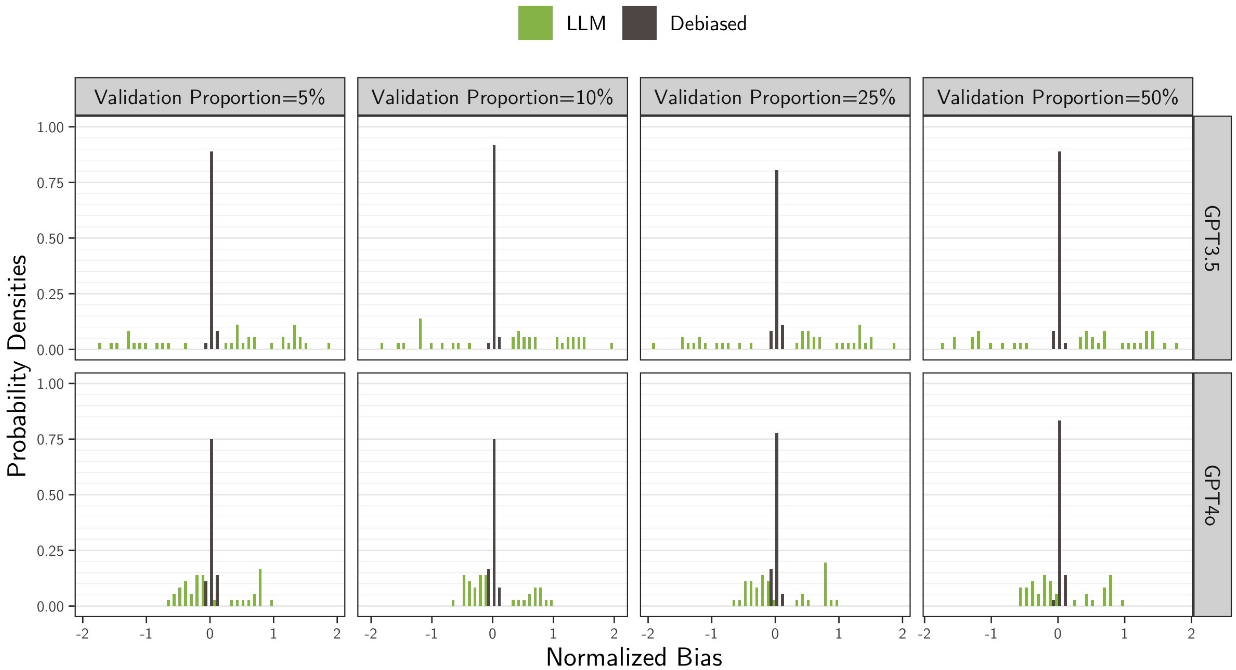


Figure 11: Normalized Bias Density of β_t, t =Banking, Finance, and Domestic Commerce, by Model and Proportion of Validation Samples. Proxy on the RHS.

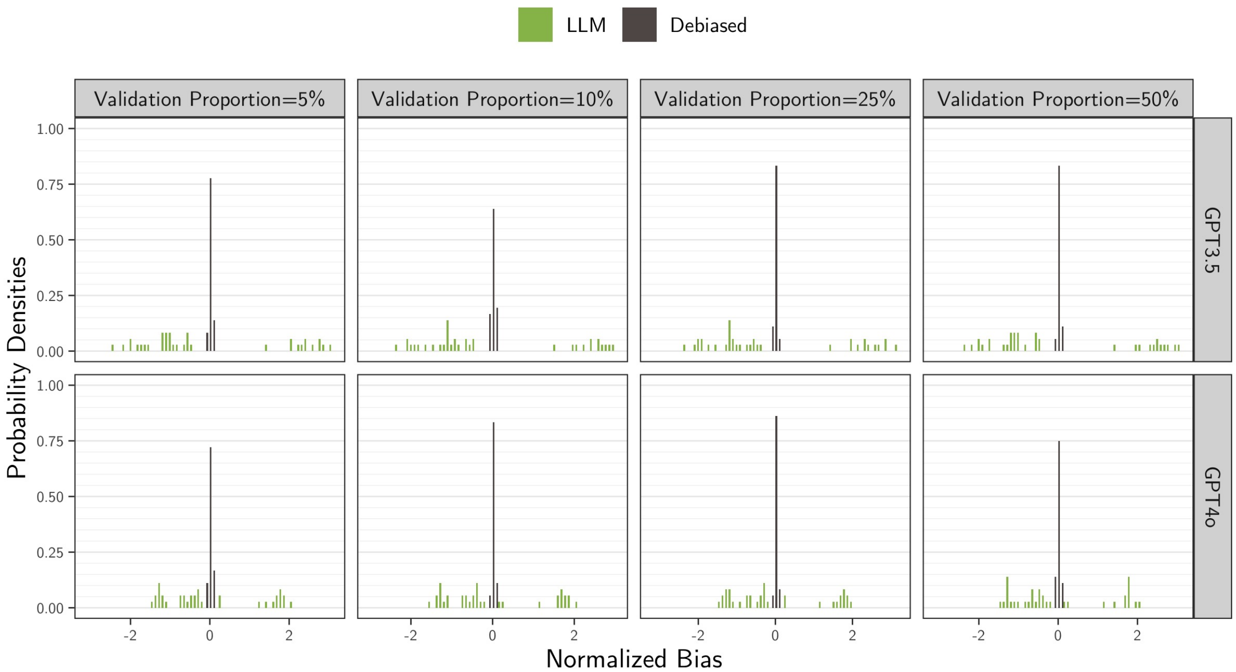


Figure 12: Normalized Bias Density of β_t, t =Defense, by Model and Proportion of Validation Samples. Proxy on the RHS.

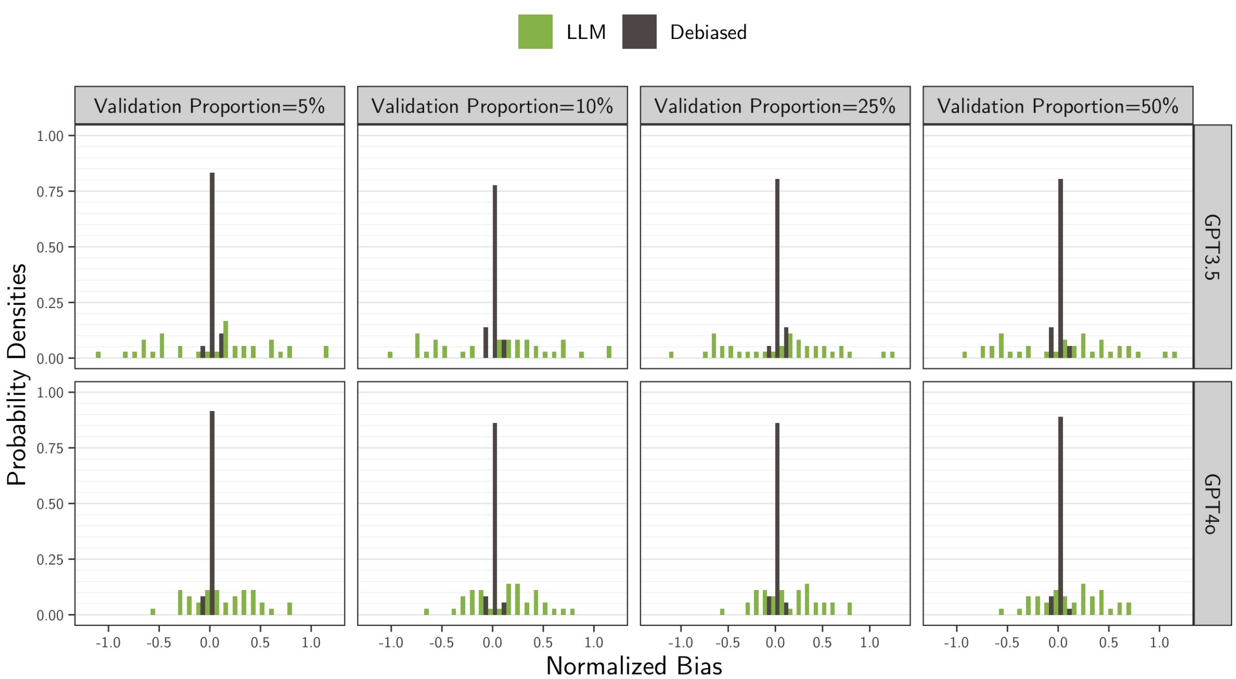


Figure 13: Normalized Bias Density of β_t, t =Government Operations, by Model and Proportion of Validation Samples. Proxy on the RHS.

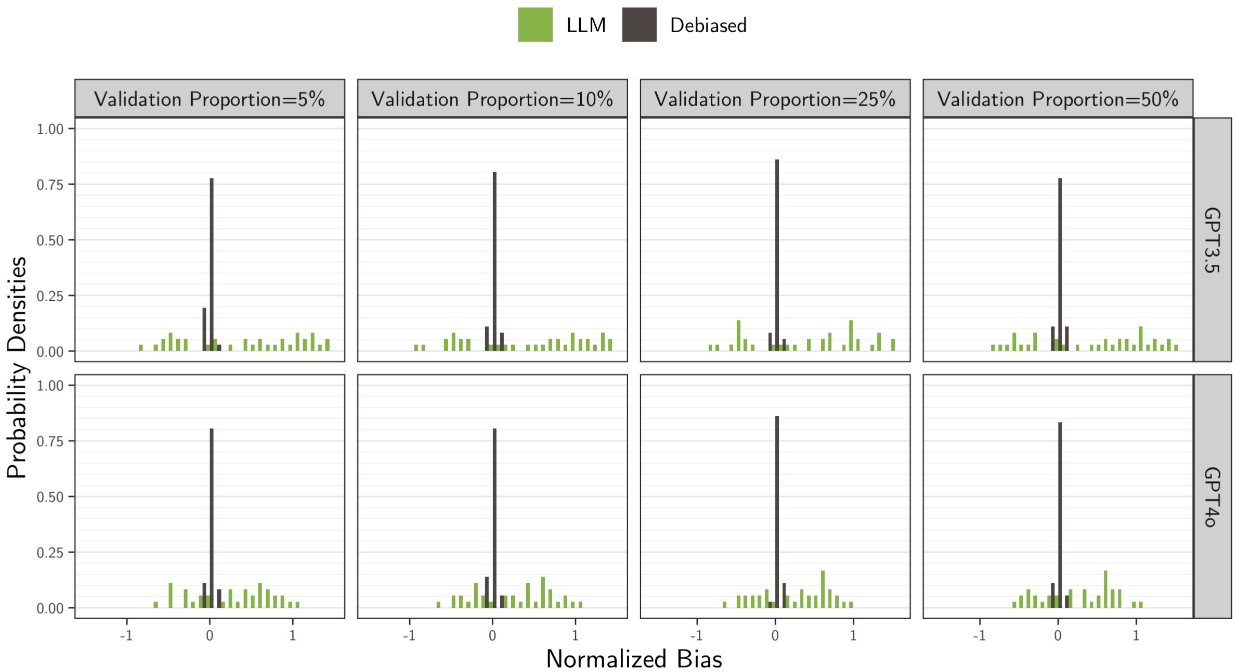


Figure 14: Normalized Bias Density of β_t, t =Public Lands and Water Management, by Model and Proportion of Validation Samples. Proxy on the RHS.

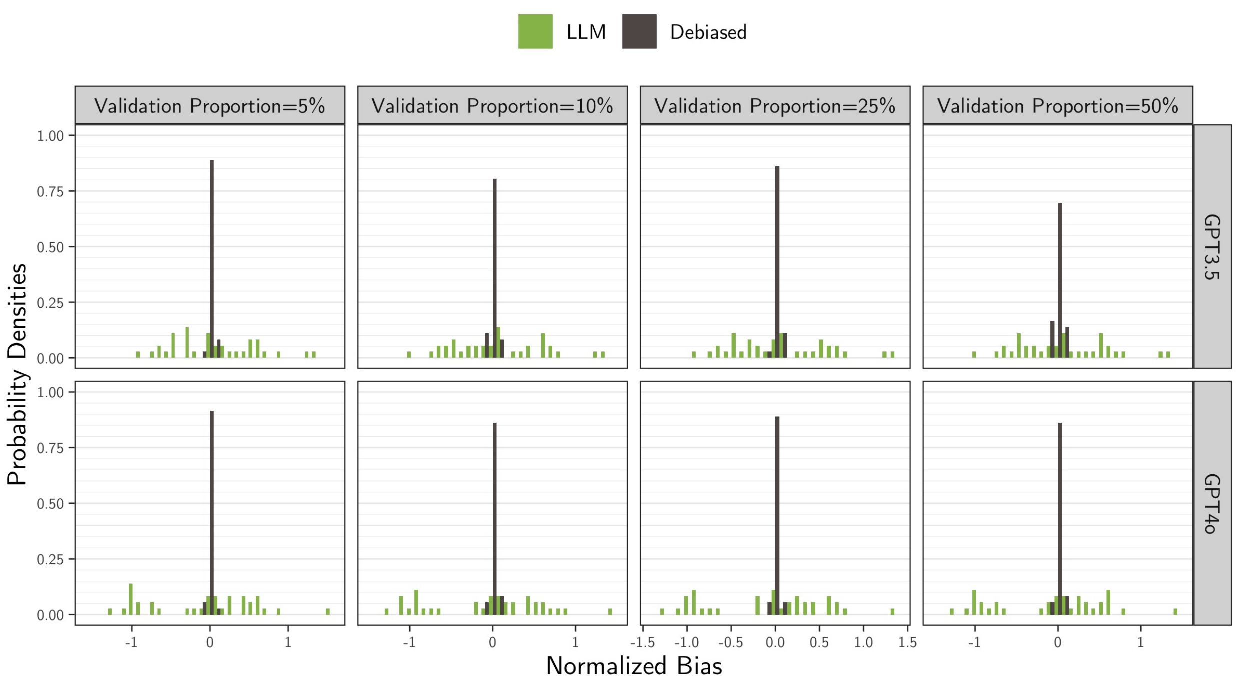


Figure 15: MSE Empirical CDF of β_t for all t by Proportion of Validation Samples. Proxy on the RHS.

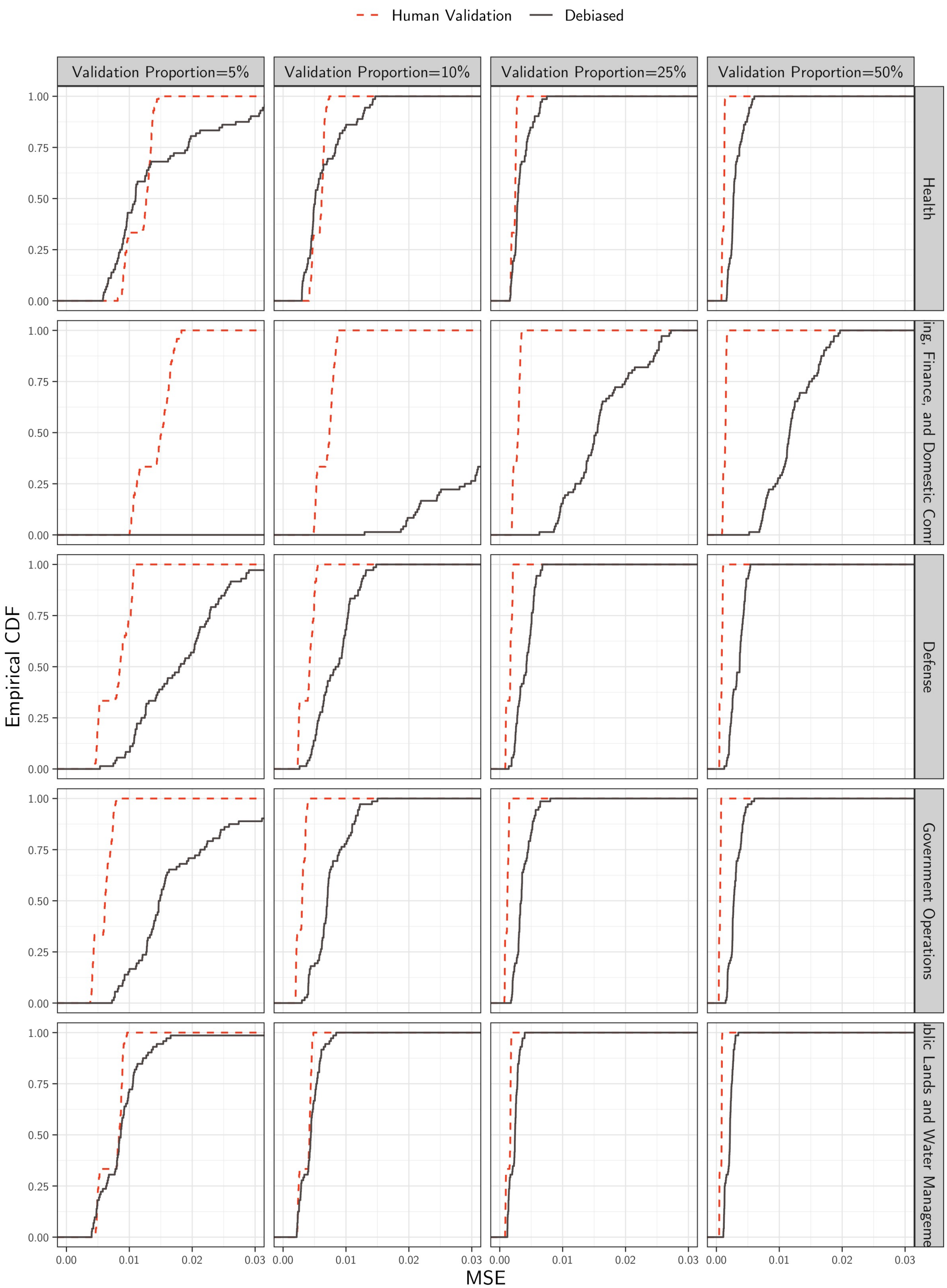


Figure 16: MSE Empirical CDF of β_t, t =Health, by Model and Proportion of Validation Samples. Proxy on the RHS.

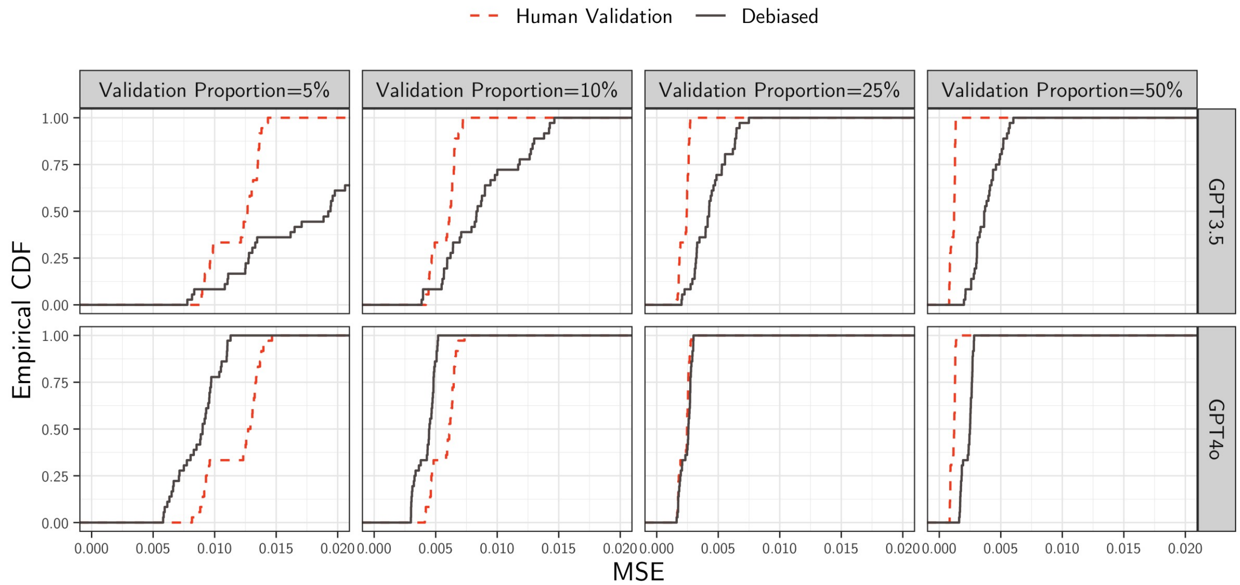


Figure 17: MSE Empirical CDF of β_t, t =Banking, Finance, and Domestic Commerce, by Model and Proportion of Validation Samples. Proxy on the RHS.

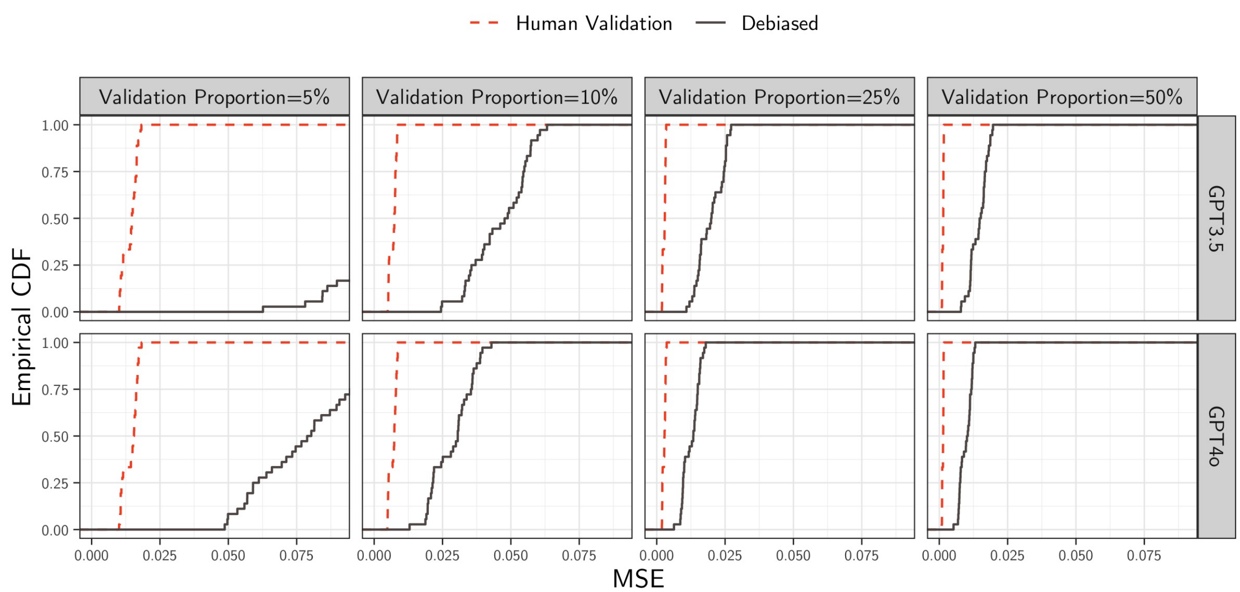


Figure 18: MSE Empirical CDF of β_t, t =Defense, by Model and Proportion of Validation Samples. Proxy on the RHS.

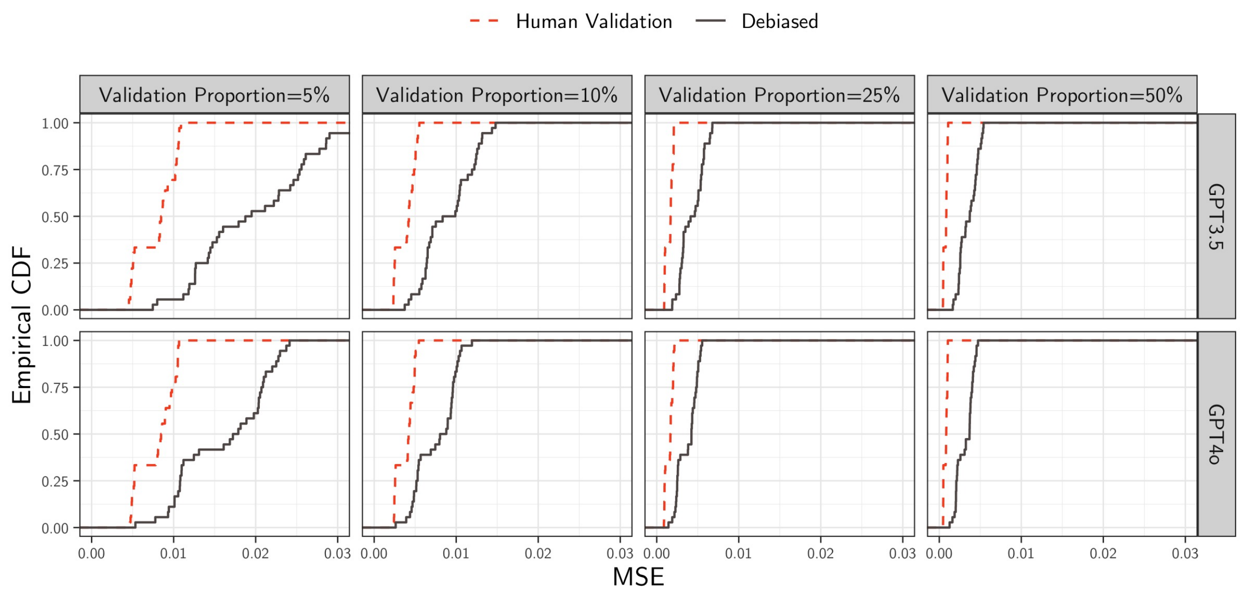


Figure 19: MSE Empirical CDF of β_t, t =Government Operations, by Model and Proportion of Validation Samples. Proxy on the RHS.

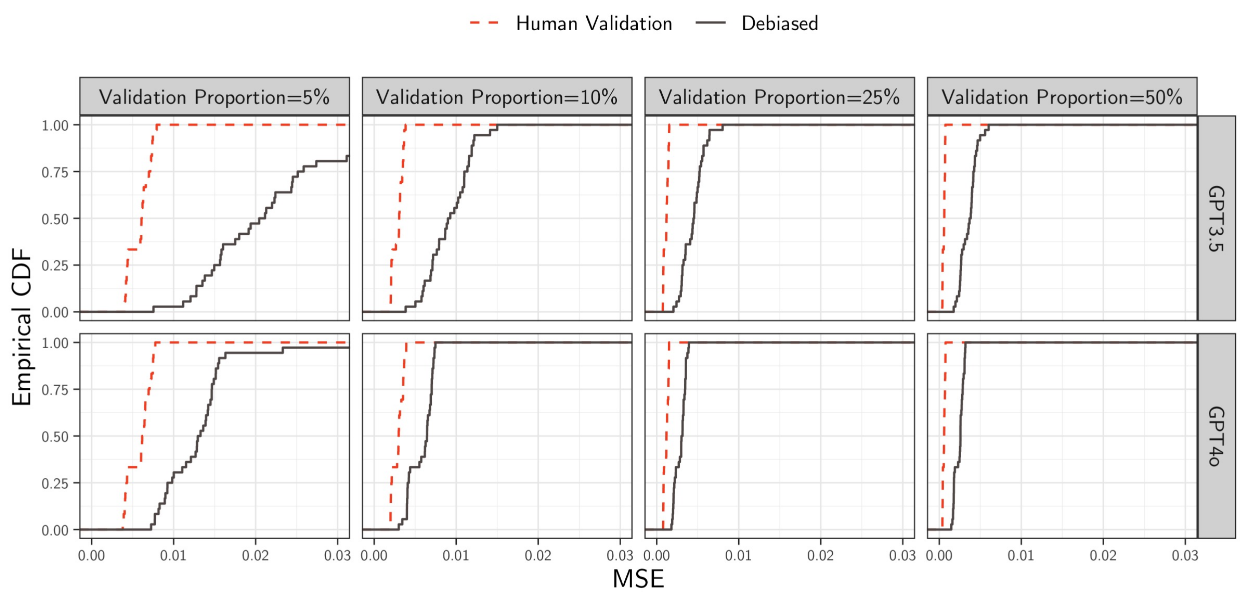


Figure 20: MSE Empirical CDF of β_t, t =Public Lands and Water Management, by Model and Proportion of Validation Samples. Proxy on the RHS.

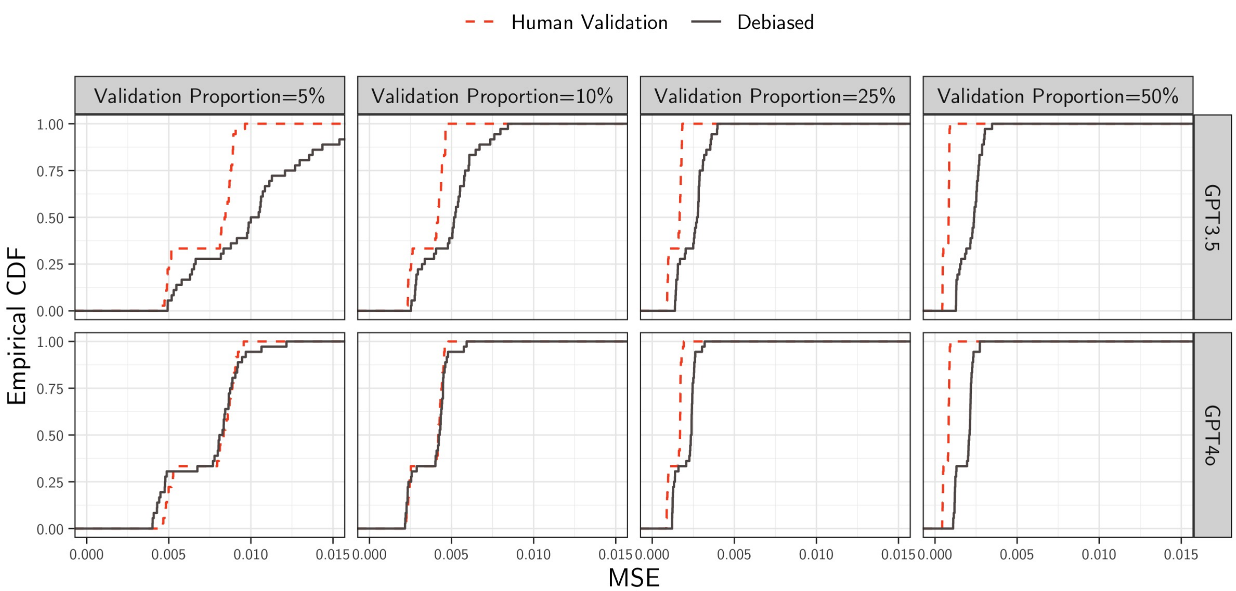


Table 1: Bias Summary Statistics of β_t, t =Health, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.003	0.021	0.004	-0.035	0.035
5%	Human Validation	0.000	0.003	0.001	-0.005	0.006
5%	Debiased	0.001	0.004	0.000	-0.005	0.006
10%	LLM	0.003	0.022	0.003	-0.035	0.035
10%	Human Validation	0.000	0.002	0.000	-0.004	0.004
10%	Debiased	0.000	0.003	0.000	-0.004	0.004
25%	LLM	0.003	0.022	0.004	-0.036	0.035
25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
25%	Debiased	0.000	0.002	0.000	-0.003	0.003
50%	LLM	0.003	0.022	0.003	-0.036	0.036
50%	Human Validation	0.000	0.001	0.000	-0.002	0.002
50%	Debiased	0.000	0.002	0.000	-0.002	0.003

Table 2: Bias Summary Statistics of β_t, t =Banking, Finance, and Domestic Commerce, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.003	0.031	-0.012	-0.043	0.049
5%	Human Validation	0.000	0.003	0.000	-0.006	0.005
5%	Debiased	0.007	0.063	0.000	-0.030	0.042
10%	LLM	-0.003	0.031	-0.012	-0.043	0.050
10%	Human Validation	0.000	0.002	-0.001	-0.004	0.003
10%	Debiased	0.001	0.008	0.001	-0.010	0.013
25%	LLM	-0.003	0.031	-0.012	-0.043	0.050
25%	Human Validation	0.000	0.002	0.000	-0.003	0.003
25%	Debiased	0.000	0.004	0.001	-0.007	0.006
50%	LLM	-0.003	0.031	-0.012	-0.043	0.048
50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
50%	Debiased	0.000	0.004	0.000	-0.005	0.006

Table 3: Bias Summary Statistics of β_t, t =Defense, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.003	0.011	0.003	-0.017	0.020
5%	Human Validation	0.000	0.003	0.000	-0.005	0.004
5%	Debiased	0.000	0.004	0.000	-0.007	0.006
10%	LLM	0.002	0.011	0.003	-0.018	0.020
10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
10%	Debiased	0.000	0.003	0.000	-0.005	0.004
25%	LLM	0.003	0.011	0.003	-0.017	0.020
25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
25%	Debiased	0.000	0.002	0.001	-0.003	0.003
50%	LLM	0.003	0.011	0.003	-0.017	0.019
50%	Human Validation	0.000	0.001	0.000	-0.002	0.001
50%	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 4: Bias Summary Statistics of β_t, t =Government Operations, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = \sum_t Y_t\beta_t$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.007	0.012	0.009	-0.011	0.023
5%	Human Validation	0.001	0.003	0.001	-0.004	0.005
5%	Debiased	-0.002	0.009	-0.001	-0.010	0.005
10%	LLM	0.007	0.012	0.009	-0.011	0.024
10%	Human Validation	0.000	0.002	0.000	-0.003	0.002
10%	Debiased	0.000	0.003	0.000	-0.005	0.004
25%	LLM	0.006	0.012	0.009	-0.011	0.024
25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
25%	Debiased	0.000	0.002	0.000	-0.003	0.003
50%	LLM	0.006	0.012	0.009	-0.011	0.024
50%	Human Validation	0.000	0.001	0.000	-0.001	0.001
50%	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 5: Bias Summary Statistics of β_t, t =Public Lands and Water Management, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = \sum_t Y_t\beta_t$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.001	0.013	0.001	-0.022	0.016
5%	Human Validation	0.000	0.003	0.000	-0.005	0.004
5%	Debiased	0.000	0.003	0.000	-0.004	0.005
10%	LLM	-0.001	0.013	0.001	-0.023	0.015
10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
10%	Debiased	0.000	0.002	0.000	-0.003	0.004
25%	LLM	-0.001	0.013	0.001	-0.021	0.015
25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
25%	Debiased	0.000	0.001	0.000	-0.002	0.003
50%	LLM	-0.001	0.012	0.001	-0.021	0.015
50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
50%	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 6: Bias Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = \sum_t Y_t\beta_t$

Validation Proportion	Bias	Mean	SD	Median	5%	95%

Table 7: Normalized Bias Summary Statistics of β_t, t =Health, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.148	0.846	0.166	-1.236	1.398
5%	Human Validation	0.001	0.032	0.005	-0.052	0.057
5%	Debiased	0.003	0.033	0.003	-0.051	0.051
10%	LLM	0.150	0.842	0.139	-1.202	1.419
10%	Human Validation	-0.002	0.029	-0.006	-0.051	0.049
10%	Debiased	0.000	0.033	-0.003	-0.056	0.050
25%	LLM	0.154	0.852	0.165	-1.287	1.419
25%	Human Validation	0.006	0.028	0.008	-0.040	0.047
25%	Debiased	-0.001	0.035	-0.001	-0.055	0.057
50%	LLM	0.150	0.845	0.124	-1.286	1.444
50%	Human Validation	0.001	0.032	0.003	-0.053	0.045
50%	Debiased	-0.002	0.029	-0.004	-0.046	0.047

Table 8: Normalized Bias Summary Statistics of β_t, t =Banking, Finance, and Domestic Commerce, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.013	1.541	-0.537	-1.885	2.673
5%	Human Validation	-0.002	0.029	-0.001	-0.050	0.044
5%	Debiased	0.004	0.039	0.002	-0.057	0.063
10%	LLM	0.013	1.543	-0.553	-1.929	2.667
10%	Human Validation	-0.006	0.028	-0.006	-0.046	0.043
10%	Debiased	0.002	0.039	0.003	-0.058	0.067
25%	LLM	0.005	1.541	-0.560	-1.934	2.622
25%	Human Validation	0.001	0.032	-0.001	-0.047	0.050
25%	Debiased	-0.001	0.034	0.005	-0.059	0.056
50%	LLM	0.009	1.547	-0.543	-1.968	2.667
50%	Human Validation	0.003	0.032	0.004	-0.042	0.050
50%	Debiased	0.001	0.034	0.001	-0.057	0.057

Table 9: Normalized Bias Summary Statistics of β_t, t =Defense, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.118	0.463	0.166	-0.633	0.791
5%	Human Validation	-0.003	0.033	-0.004	-0.056	0.050
5%	Debiased	-0.002	0.030	-0.002	-0.046	0.045
10%	LLM	0.108	0.470	0.167	-0.695	0.764
10%	Human Validation	-0.001	0.034	0.000	-0.047	0.056
10%	Debiased	-0.003	0.033	-0.002	-0.060	0.048
25%	LLM	0.121	0.460	0.150	-0.653	0.782
25%	Human Validation	-0.005	0.034	-0.012	-0.055	0.056
25%	Debiased	0.004	0.035	0.007	-0.052	0.054
50%	LLM	0.112	0.450	0.158	-0.649	0.762
50%	Human Validation	-0.006	0.028	-0.003	-0.052	0.035
50%	Debiased	-0.002	0.030	0.000	-0.051	0.042

Table 10: Normalized Bias Summary Statistics of β_t, t =Government Operations, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.369	0.615	0.486	-0.534	1.285
5%	Human Validation	0.008	0.034	0.010	-0.050	0.062
5%	Debiased	-0.010	0.034	-0.009	-0.064	0.039
10%	LLM	0.368	0.615	0.466	-0.502	1.344
10%	Human Validation	-0.004	0.029	-0.006	-0.044	0.039
10%	Debiased	-0.004	0.034	-0.003	-0.064	0.048
25%	LLM	0.359	0.607	0.451	-0.493	1.326
25%	Human Validation	-0.001	0.032	-0.001	-0.058	0.055
25%	Debiased	0.000	0.033	-0.002	-0.045	0.054
50%	LLM	0.361	0.619	0.466	-0.554	1.331
50%	Human Validation	0.007	0.027	0.007	-0.034	0.047
50%	Debiased	-0.002	0.035	0.000	-0.069	0.049

Table 11: Normalized Bias Summary Statistics of β_t, t =Public Lands and Water Management, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.008	0.632	0.023	-1.000	0.888
5%	Human Validation	-0.001	0.034	0.001	-0.059	0.047
5%	Debiased	0.002	0.030	0.001	-0.041	0.048
10%	LLM	-0.002	0.630	0.041	-1.005	0.864
10%	Human Validation	0.001	0.030	0.002	-0.049	0.043
10%	Debiased	0.004	0.033	0.004	-0.048	0.056
25%	LLM	-0.009	0.623	0.029	-0.970	0.838
25%	Human Validation	0.003	0.030	0.004	-0.049	0.047
25%	Debiased	-0.001	0.030	-0.007	-0.040	0.060
50%	LLM	-0.003	0.619	0.049	-0.990	0.835
50%	Human Validation	0.001	0.034	0.000	-0.055	0.054
50%	Debiased	0.000	0.036	0.003	-0.055	0.061

Table 12: Normalized Bias Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%

Table 13: Bias Summary Statistics of β_t, t =Health, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.004	0.028	0.013	-0.042	0.037
GPT-3.5	5%	Human Validation	0.001	0.004	0.001	-0.005	0.006
GPT-3.5	5%	Debiased	0.001	0.005	0.000	-0.005	0.007
GPT-3.5	10%	LLM	0.004	0.029	0.012	-0.041	0.038
GPT-3.5	10%	Human Validation	0.000	0.002	0.000	-0.004	0.004
GPT-3.5	10%	Debiased	0.000	0.003	0.000	-0.003	0.004
GPT-3.5	25%	LLM	0.004	0.029	0.012	-0.041	0.038
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-3.5	25%	Debiased	0.000	0.002	0.000	-0.003	0.004
GPT-3.5	50%	LLM	0.004	0.029	0.012	-0.041	0.037
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-3.5	50%	Debiased	0.000	0.002	0.000	-0.003	0.003
GPT-4o	5%	LLM	0.001	0.011	-0.003	-0.012	0.017
GPT-4o	5%	Human Validation	-0.001	0.003	0.000	-0.005	0.004
GPT-4o	5%	Debiased	0.000	0.003	0.000	-0.005	0.005
GPT-4o	10%	LLM	0.001	0.011	-0.003	-0.012	0.017
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
GPT-4o	10%	Debiased	0.000	0.002	0.000	-0.004	0.004
GPT-4o	25%	LLM	0.001	0.011	-0.003	-0.012	0.017
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.002	0.003
GPT-4o	25%	Debiased	0.000	0.002	0.000	-0.003	0.003
GPT-4o	50%	LLM	0.001	0.011	-0.003	-0.012	0.016
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-4o	50%	Debiased	0.000	0.002	0.000	-0.002	0.003

Table 14: Bias Summary Statistics of β_t, t =Banking, Finance, and Domestic Commerce, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-0.004	0.036	-0.018	-0.044	0.051
GPT-3.5	5%	Human Validation	-0.001	0.004	-0.001	-0.006	0.004
GPT-3.5	5%	Debiased	0.012	0.086	0.000	-0.034	0.051
GPT-3.5	10%	LLM	-0.003	0.036	-0.019	-0.045	0.052
GPT-3.5	10%	Human Validation	-0.001	0.002	-0.001	-0.004	0.003
GPT-3.5	10%	Debiased	0.001	0.009	0.002	-0.012	0.013
GPT-3.5	25%	LLM	-0.004	0.036	-0.017	-0.046	0.052
GPT-3.5	25%	Human Validation	0.000	0.002	0.000	-0.002	0.002
GPT-3.5	25%	Debiased	0.000	0.005	0.001	-0.008	0.006
GPT-3.5	50%	LLM	-0.004	0.036	-0.020	-0.045	0.051
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
GPT-3.5	50%	Debiased	0.001	0.004	0.000	-0.005	0.006
GPT-4o	5%	LLM	-0.002	0.025	-0.008	-0.031	0.036
GPT-4o	5%	Human Validation	0.000	0.003	0.000	-0.005	0.005
GPT-4o	5%	Debiased	0.002	0.021	0.001	-0.021	0.032
GPT-4o	10%	LLM	-0.002	0.025	-0.009	-0.032	0.036
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.004	0.003
GPT-4o	10%	Debiased	0.000	0.006	0.000	-0.009	0.012
GPT-4o	25%	LLM	-0.002	0.025	-0.008	-0.032	0.036
GPT-4o	25%	Human Validation	0.000	0.002	0.000	-0.003	0.003
GPT-4o	25%	Debiased	0.000	0.004	0.000	-0.005	0.006
GPT-4o	50%	LLM	-0.002	0.025	-0.009	-0.032	0.035
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	50%	Debiased	0.000	0.004	0.000	-0.005	0.005

Table 15: Bias Summary Statistics of β_t, t =Defense, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.002	0.013	0.003	-0.018	0.023
GPT-3.5	5%	Human Validation	0.000	0.003	-0.001	-0.004	0.005
GPT-3.5	5%	Debiased	0.000	0.005	0.001	-0.006	0.007
GPT-3.5	10%	LLM	0.001	0.014	0.003	-0.018	0.022
GPT-3.5	10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
GPT-3.5	10%	Debiased	0.000	0.003	0.000	-0.005	0.004
GPT-3.5	25%	LLM	0.002	0.014	0.003	-0.017	0.022
GPT-3.5	25%	Human Validation	0.000	0.001	-0.001	-0.002	0.002
GPT-3.5	25%	Debiased	0.000	0.003	0.000	-0.003	0.004
GPT-3.5	50%	LLM	0.001	0.013	0.003	-0.017	0.022
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-3.5	50%	Debiased	0.000	0.002	0.000	-0.003	0.003
GPT-4o	5%	LLM	0.004	0.008	0.003	-0.008	0.016
GPT-4o	5%	Human Validation	-0.001	0.003	0.000	-0.005	0.004
GPT-4o	5%	Debiased	-0.001	0.004	-0.001	-0.007	0.004
GPT-4o	10%	LLM	0.003	0.008	0.004	-0.008	0.016
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.003	0.004
GPT-4o	10%	Debiased	-0.001	0.003	0.000	-0.004	0.003
GPT-4o	25%	LLM	0.004	0.008	0.003	-0.007	0.017
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.002	0.003
GPT-4o	25%	Debiased	0.000	0.002	0.001	-0.004	0.003
GPT-4o	50%	LLM	0.004	0.008	0.003	-0.008	0.017
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-4o	50%	Debiased	0.000	0.002	0.000	-0.003	0.002

Table 16: Bias Summary Statistics of β_t, t =Government Operations, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.008	0.015	0.013	-0.013	0.024
GPT-3.5	5%	Human Validation	0.000	0.003	0.000	-0.005	0.005
GPT-3.5	5%	Debiased	-0.003	0.012	-0.001	-0.012	0.005
GPT-3.5	10%	LLM	0.009	0.015	0.014	-0.013	0.024
GPT-3.5	10%	Human Validation	0.000	0.002	0.000	-0.003	0.002
GPT-3.5	10%	Debiased	-0.001	0.003	-0.001	-0.006	0.004
GPT-3.5	25%	LLM	0.008	0.015	0.014	-0.013	0.025
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.001	0.002
GPT-3.5	25%	Debiased	0.000	0.002	0.000	-0.004	0.003
GPT-3.5	50%	LLM	0.008	0.015	0.013	-0.013	0.025
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	50%	Debiased	0.000	0.002	0.000	-0.004	0.003
GPT-4o	5%	LLM	0.005	0.008	0.007	-0.008	0.016
GPT-4o	5%	Human Validation	0.001	0.002	0.001	-0.003	0.005
GPT-4o	5%	Debiased	-0.001	0.004	-0.001	-0.007	0.005
GPT-4o	10%	LLM	0.005	0.008	0.007	-0.008	0.017
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.003	0.002
GPT-4o	10%	Debiased	0.000	0.003	0.001	-0.005	0.004
GPT-4o	25%	LLM	0.004	0.008	0.006	-0.008	0.015
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	25%	Debiased	0.000	0.002	0.000	-0.002	0.003
GPT-4o	50%	LLM	0.004	0.008	0.006	-0.009	0.016
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-4o	50%	Debiased	0.000	0.002	0.000	-0.003	0.002

Table 17: Bias Summary Statistics of β_t, t =Public Lands and Water Management, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.011	0.001	-0.015	0.018
GPT-3.5	5%	Human Validation	0.000	0.003	0.000	-0.006	0.005
GPT-3.5	5%	Debiased	0.001	0.003	0.001	-0.004	0.007
GPT-3.5	10%	LLM	0.001	0.011	0.001	-0.015	0.017
GPT-3.5	10%	Human Validation	0.000	0.002	0.000	-0.002	0.003
GPT-3.5	10%	Debiased	0.000	0.003	0.000	-0.004	0.004
GPT-3.5	25%	LLM	0.001	0.011	0.000	-0.015	0.017
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.001	0.002
GPT-3.5	25%	Debiased	0.000	0.002	0.000	-0.002	0.003
GPT-3.5	50%	LLM	0.001	0.011	0.001	-0.015	0.017
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
GPT-3.5	50%	Debiased	0.000	0.002	0.000	-0.003	0.003
GPT-4o	5%	LLM	-0.003	0.014	0.001	-0.022	0.012
GPT-4o	5%	Human Validation	0.000	0.003	0.000	-0.004	0.004
GPT-4o	5%	Debiased	-0.001	0.002	-0.001	-0.004	0.003
GPT-4o	10%	LLM	-0.003	0.014	0.001	-0.023	0.013
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.003	0.002
GPT-4o	10%	Debiased	0.000	0.002	0.001	-0.003	0.003
GPT-4o	25%	LLM	-0.003	0.014	0.001	-0.022	0.012
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	25%	Debiased	0.000	0.001	0.000	-0.002	0.001
GPT-4o	50%	LLM	-0.003	0.014	0.001	-0.022	0.012
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	50%	Debiased	0.000	0.001	0.000	-0.002	0.002

Table 18: Bias Summary Statistics of β_{Other} by Model and Proportion of Validation Samples for. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%

Table 19: Normalized Bias Summary of β_t, t =Health, Statistics by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.212	1.089	0.483	-1.480	1.463
GPT-3.5	5%	Human Validation	0.008	0.034	0.006	-0.049	0.063
GPT-3.5	5%	Debiased	0.001	0.028	0.001	-0.042	0.048
GPT-3.5	10%	LLM	0.225	1.085	0.500	-1.453	1.500
GPT-3.5	10%	Human Validation	0.000	0.030	-0.005	-0.051	0.055
GPT-3.5	10%	Debiased	0.005	0.028	0.006	-0.032	0.046
GPT-3.5	25%	LLM	0.230	1.094	0.538	-1.462	1.493
GPT-3.5	25%	Human Validation	0.006	0.025	0.011	-0.031	0.041
GPT-3.5	25%	Debiased	0.002	0.034	0.001	-0.045	0.057
GPT-3.5	50%	LLM	0.222	1.091	0.515	-1.490	1.519
GPT-3.5	50%	Human Validation	-0.001	0.034	-0.005	-0.055	0.051
GPT-3.5	50%	Debiased	-0.006	0.027	-0.004	-0.050	0.038
GPT-4o	5%	LLM	0.084	0.507	-0.119	-0.510	0.824
GPT-4o	5%	Human Validation	-0.005	0.029	-0.003	-0.051	0.037
GPT-4o	5%	Debiased	0.005	0.037	0.005	-0.067	0.055
GPT-4o	10%	LLM	0.075	0.501	-0.135	-0.468	0.825
GPT-4o	10%	Human Validation	-0.005	0.028	-0.007	-0.043	0.042
GPT-4o	10%	Debiased	-0.006	0.036	-0.006	-0.074	0.052
GPT-4o	25%	LLM	0.078	0.513	-0.136	-0.483	0.857
GPT-4o	25%	Human Validation	0.006	0.032	0.006	-0.048	0.055
GPT-4o	25%	Debiased	-0.005	0.036	-0.001	-0.061	0.048
GPT-4o	50%	LLM	0.078	0.497	-0.119	-0.514	0.805
GPT-4o	50%	Human Validation	0.003	0.029	0.013	-0.041	0.038
GPT-4o	50%	Debiased	0.001	0.032	-0.004	-0.041	0.066

Table 20: Normalized Bias Summary Statistics of β_t, t =Banking, Finance, and Domestic Commerce, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-0.003	1.821	-0.848	-2.053	2.819
GPT-3.5	5%	Human Validation	-0.006	0.030	-0.006	-0.053	0.032
GPT-3.5	5%	Debiased	0.003	0.035	0.000	-0.057	0.061
GPT-3.5	10%	LLM	0.003	1.826	-0.853	-2.059	2.826
GPT-3.5	10%	Human Validation	-0.012	0.028	-0.012	-0.048	0.039
GPT-3.5	10%	Debiased	0.002	0.043	0.009	-0.062	0.061
GPT-3.5	25%	LLM	-0.021	1.822	-0.808	-2.035	2.855
GPT-3.5	25%	Human Validation	0.002	0.029	0.002	-0.035	0.044
GPT-3.5	25%	Debiased	-0.001	0.034	0.006	-0.068	0.039
GPT-3.5	50%	LLM	-0.003	1.832	-0.886	-2.059	2.815
GPT-3.5	50%	Human Validation	0.012	0.031	0.011	-0.039	0.053
GPT-3.5	50%	Debiased	0.005	0.031	0.001	-0.042	0.053
GPT-4o	5%	LLM	0.028	1.224	-0.369	-1.314	1.908
GPT-4o	5%	Human Validation	0.003	0.028	0.004	-0.043	0.045
GPT-4o	5%	Debiased	0.006	0.043	0.004	-0.056	0.067
GPT-4o	10%	LLM	0.023	1.224	-0.370	-1.344	1.890
GPT-4o	10%	Human Validation	0.000	0.027	0.001	-0.040	0.044
GPT-4o	10%	Debiased	0.001	0.035	0.002	-0.046	0.073
GPT-4o	25%	LLM	0.031	1.222	-0.332	-1.323	1.879
GPT-4o	25%	Human Validation	0.000	0.036	-0.003	-0.057	0.053
GPT-4o	25%	Debiased	0.000	0.033	0.000	-0.043	0.056
GPT-4o	50%	LLM	0.022	1.223	-0.414	-1.309	1.859
GPT-4o	50%	Human Validation	-0.006	0.030	-0.010	-0.043	0.047
GPT-4o	50%	Debiased	-0.004	0.038	0.000	-0.061	0.055

Table 21: Normalized Bias Summary Statistics of β_t, t =Defense, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.075	0.568	0.189	-0.712	0.900
GPT-3.5	5%	Human Validation	0.000	0.033	-0.007	-0.052	0.056
GPT-3.5	5%	Debiased	0.002	0.032	0.004	-0.039	0.049
GPT-3.5	10%	LLM	0.069	0.577	0.167	-0.725	0.950
GPT-3.5	10%	Human Validation	-0.002	0.035	0.002	-0.041	0.047
GPT-3.5	10%	Debiased	-0.001	0.035	-0.002	-0.062	0.048
GPT-3.5	25%	LLM	0.073	0.566	0.150	-0.688	0.930
GPT-3.5	25%	Human Validation	-0.011	0.031	-0.015	-0.056	0.043
GPT-3.5	25%	Debiased	0.005	0.039	0.005	-0.046	0.061
GPT-3.5	50%	LLM	0.072	0.550	0.132	-0.683	0.902
GPT-3.5	50%	Human Validation	-0.007	0.025	-0.002	-0.048	0.032
GPT-3.5	50%	Debiased	-0.004	0.031	-0.007	-0.051	0.045
GPT-4o	5%	LLM	0.160	0.328	0.140	-0.276	0.693
GPT-4o	5%	Human Validation	-0.006	0.033	0.000	-0.055	0.048
GPT-4o	5%	Debiased	-0.006	0.028	-0.006	-0.050	0.033
GPT-4o	10%	LLM	0.148	0.334	0.171	-0.283	0.664
GPT-4o	10%	Human Validation	0.001	0.033	-0.003	-0.049	0.064
GPT-4o	10%	Debiased	-0.005	0.030	-0.002	-0.055	0.037
GPT-4o	25%	LLM	0.169	0.322	0.152	-0.241	0.685
GPT-4o	25%	Human Validation	0.001	0.037	-0.006	-0.045	0.071
GPT-4o	25%	Debiased	0.004	0.031	0.010	-0.056	0.044
GPT-4o	50%	LLM	0.152	0.324	0.158	-0.300	0.654
GPT-4o	50%	Human Validation	-0.005	0.032	-0.005	-0.055	0.038
GPT-4o	50%	Debiased	0.000	0.030	0.003	-0.054	0.040

Table 22: Normalized Bias Summary Statistics of β_t, $t =$Government Operations, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.448	0.723	0.594	-0.586	1.374
GPT-3.5	5%	Human Validation	0.004	0.038	0.004	-0.059	0.064
GPT-3.5	5%	Debiased	-0.009	0.035	-0.008	-0.062	0.033
GPT-3.5	10%	LLM	0.452	0.728	0.665	-0.575	1.387
GPT-3.5	10%	Human Validation	-0.003	0.030	-0.007	-0.046	0.034
GPT-3.5	10%	Debiased	-0.007	0.034	-0.008	-0.061	0.048
GPT-3.5	25%	LLM	0.443	0.723	0.642	-0.572	1.402
GPT-3.5	25%	Human Validation	-0.001	0.030	-0.002	-0.043	0.062
GPT-3.5	25%	Debiased	-0.002	0.034	-0.004	-0.051	0.046
GPT-3.5	50%	LLM	0.447	0.737	0.622	-0.620	1.413
GPT-3.5	50%	Human Validation	0.008	0.026	0.008	-0.032	0.049
GPT-3.5	50%	Debiased	0.002	0.036	0.005	-0.058	0.052
GPT-4o	5%	LLM	0.290	0.482	0.423	-0.435	0.905
GPT-4o	5%	Human Validation	0.011	0.030	0.015	-0.043	0.054
GPT-4o	5%	Debiased	-0.010	0.033	-0.011	-0.064	0.047
GPT-4o	10%	LLM	0.285	0.471	0.412	-0.420	0.914
GPT-4o	10%	Human Validation	-0.004	0.028	-0.006	-0.043	0.041
GPT-4o	10%	Debiased	-0.001	0.035	0.010	-0.064	0.046
GPT-4o	25%	LLM	0.276	0.458	0.392	-0.441	0.836
GPT-4o	25%	Human Validation	-0.001	0.033	0.000	-0.061	0.050
GPT-4o	25%	Debiased	0.003	0.032	0.000	-0.035	0.060
GPT-4o	50%	LLM	0.275	0.468	0.382	-0.447	0.859
GPT-4o	50%	Human Validation	0.005	0.029	0.005	-0.034	0.041
GPT-4o	50%	Debiased	-0.006	0.033	-0.005	-0.068	0.041

Table 23: Normalized Bias Summary Statistics of β_t, t =Public Lands and Water Management, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.068	0.557	0.023	-0.676	0.977
GPT-3.5	5%	Human Validation	0.001	0.038	0.003	-0.070	0.052
GPT-3.5	5%	Debiased	0.009	0.030	0.012	-0.040	0.053
GPT-3.5	10%	LLM	0.072	0.558	0.040	-0.657	0.938
GPT-3.5	10%	Human Validation	0.005	0.032	0.001	-0.038	0.055
GPT-3.5	10%	Debiased	0.000	0.036	0.001	-0.053	0.059
GPT-3.5	25%	LLM	0.070	0.561	0.012	-0.680	0.950
GPT-3.5	25%	Human Validation	0.012	0.028	0.011	-0.031	0.057
GPT-3.5	25%	Debiased	0.002	0.032	-0.008	-0.037	0.066
GPT-3.5	50%	LLM	0.076	0.554	0.052	-0.673	0.943
GPT-3.5	50%	Human Validation	0.004	0.029	-0.002	-0.037	0.053
GPT-3.5	50%	Debiased	-0.001	0.038	0.003	-0.059	0.059
GPT-4o	5%	LLM	-0.085	0.697	0.038	-1.051	0.757
GPT-4o	5%	Human Validation	-0.003	0.031	0.001	-0.050	0.042
GPT-4o	5%	Debiased	-0.006	0.028	-0.010	-0.039	0.041
GPT-4o	10%	LLM	-0.076	0.695	0.041	-1.062	0.803
GPT-4o	10%	Human Validation	-0.003	0.028	0.002	-0.052	0.032
GPT-4o	10%	Debiased	0.008	0.029	0.010	-0.041	0.050
GPT-4o	25%	LLM	-0.087	0.677	0.042	-1.030	0.752
GPT-4o	25%	Human Validation	-0.005	0.031	0.001	-0.057	0.044
GPT-4o	25%	Debiased	-0.003	0.029	-0.007	-0.044	0.041
GPT-4o	50%	LLM	-0.083	0.677	0.047	-1.015	0.699
GPT-4o	50%	Human Validation	-0.001	0.038	0.003	-0.068	0.053
GPT-4o	50%	Debiased	0.001	0.033	0.001	-0.045	0.059

Table 24: Normalized Bias Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%

Table 25: MSE Summary Statistics of β_t, t =Health, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.001	0.001	0.002
5%	Human Validation	0.012	0.002	0.013	0.009	0.014
5%	Debiased	0.015	0.015	0.011	0.006	0.031
10%	LLM	0.001	0.001	0.001	0.001	0.002
10%	Human Validation	0.006	0.001	0.006	0.004	0.007
10%	Debiased	0.007	0.003	0.005	0.003	0.013
25%	LLM	0.001	0.001	0.001	0.001	0.003
25%	Human Validation	0.002	0.000	0.002	0.002	0.003
25%	Debiased	0.003	0.001	0.003	0.002	0.006
50%	LLM	0.001	0.001	0.001	0.001	0.002
50%	Human Validation	0.001	0.000	0.001	0.001	0.001
50%	Debiased	0.003	0.001	0.003	0.002	0.005

Table 26: MSE Summary Statistics of β_t, t =Banking, Finance, and Domestic Commerce, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.001	0.001	0.003
5%	Human Validation	0.014	0.003	0.015	0.010	0.017
5%	Debiased	3.543	23.671	0.099	0.055	2.287
10%	LLM	0.001	0.001	0.001	0.001	0.003
10%	Human Validation	0.007	0.001	0.007	0.005	0.009
10%	Debiased	0.037	0.013	0.036	0.020	0.057
25%	LLM	0.001	0.001	0.001	0.001	0.003
25%	Human Validation	0.003	0.001	0.003	0.002	0.003
25%	Debiased	0.016	0.005	0.015	0.009	0.026
50%	LLM	0.001	0.001	0.001	0.001	0.003
50%	Human Validation	0.001	0.000	0.001	0.001	0.002
50%	Debiased	0.012	0.004	0.012	0.007	0.018

Table 27: MSE Summary Statistics of β_t, t =Defense, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.001	0.000	0.001
5%	Human Validation	0.008	0.002	0.009	0.005	0.011
5%	Debiased	0.018	0.006	0.018	0.009	0.029
10%	LLM	0.001	0.000	0.001	0.000	0.001
10%	Human Validation	0.004	0.001	0.004	0.002	0.005
10%	Debiased	0.008	0.003	0.009	0.004	0.013
25%	LLM	0.001	0.000	0.001	0.000	0.001
25%	Human Validation	0.002	0.000	0.002	0.001	0.002
25%	Debiased	0.004	0.001	0.004	0.002	0.006
50%	LLM	0.001	0.000	0.001	0.000	0.001
50%	Human Validation	0.001	0.000	0.001	0.000	0.001
50%	Debiased	0.003	0.001	0.004	0.002	0.005

Table 28: MSE Summary Statistics of β_t, t =Government Operations, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.001	0.000	0.001
5%	Human Validation	0.006	0.001	0.006	0.004	0.008
5%	Debiased	0.061	0.369	0.015	0.008	0.038
10%	LLM	0.001	0.000	0.001	0.000	0.001
10%	Human Validation	0.003	0.001	0.003	0.002	0.004
10%	Debiased	0.008	0.003	0.007	0.004	0.012
25%	LLM	0.001	0.000	0.001	0.000	0.001
25%	Human Validation	0.001	0.000	0.001	0.001	0.001
25%	Debiased	0.004	0.001	0.003	0.002	0.006
50%	LLM	0.001	0.000	0.001	0.000	0.001
50%	Human Validation	0.001	0.000	0.001	0.000	0.001
50%	Debiased	0.003	0.001	0.003	0.002	0.005

Table 29: MSE Summary Statistics of β_t, t =Public Lands and Water Management, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.001	0.000	0.001
5%	Human Validation	0.007	0.002	0.008	0.005	0.009
5%	Debiased	0.009	0.006	0.009	0.004	0.015
10%	LLM	0.001	0.000	0.001	0.000	0.001
10%	Human Validation	0.004	0.001	0.004	0.002	0.005
10%	Debiased	0.004	0.001	0.004	0.002	0.007
25%	LLM	0.001	0.000	0.001	0.000	0.001
25%	Human Validation	0.001	0.000	0.002	0.001	0.002
25%	Debiased	0.002	0.001	0.002	0.001	0.004
50%	LLM	0.001	0.000	0.001	0.000	0.001
50%	Human Validation	0.001	0.000	0.001	0.000	0.001
50%	Debiased	0.002	0.001	0.002	0.001	0.003

Table 30: MSE Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	MSE	Mean	SD	Median	5%	95%

Table 31: MSE Summary Statistics of β_t, t =Health, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.001	0.001	0.001	0.003
GPT-3.5	5%	Human Validation	0.012	0.002	0.013	0.009	0.014
GPT-3.5	5%	Debiased	0.022	0.019	0.019	0.008	0.032
GPT-3.5	10%	LLM	0.001	0.001	0.002	0.001	0.003
GPT-3.5	10%	Human Validation	0.006	0.001	0.006	0.004	0.007
GPT-3.5	10%	Debiased	0.009	0.003	0.008	0.004	0.014
GPT-3.5	25%	LLM	0.001	0.001	0.001	0.001	0.003
GPT-3.5	25%	Human Validation	0.002	0.000	0.002	0.002	0.003
GPT-3.5	25%	Debiased	0.004	0.001	0.004	0.002	0.007
GPT-3.5	50%	LLM	0.001	0.001	0.002	0.001	0.003
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-3.5	50%	Debiased	0.004	0.001	0.004	0.002	0.006
GPT-4o	5%	LLM	0.001	0.000	0.001	0.001	0.001
GPT-4o	5%	Human Validation	0.012	0.002	0.013	0.009	0.014
GPT-4o	5%	Debiased	0.009	0.002	0.009	0.006	0.011
GPT-4o	10%	LLM	0.001	0.000	0.001	0.001	0.001
GPT-4o	10%	Human Validation	0.006	0.001	0.006	0.004	0.007
GPT-4o	10%	Debiased	0.004	0.001	0.004	0.003	0.005
GPT-4o	25%	LLM	0.001	0.000	0.001	0.001	0.001
GPT-4o	25%	Human Validation	0.002	0.000	0.002	0.002	0.003
GPT-4o	25%	Debiased	0.002	0.000	0.003	0.002	0.003
GPT-4o	50%	LLM	0.001	0.000	0.001	0.001	0.001
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-4o	50%	Debiased	0.002	0.000	0.002	0.002	0.003

Table 32: MSE Summary Statistics of β_t, t =Banking, Finance, and Domestic Commerce, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.002	0.001	0.002	0.001	0.003
GPT-3.5	5%	Human Validation	0.014	0.003	0.015	0.010	0.018
GPT-3.5	5%	Debiased	6.812	33.377	0.160	0.083	11.343
GPT-3.5	10%	LLM	0.002	0.001	0.002	0.001	0.003
GPT-3.5	10%	Human Validation	0.007	0.001	0.007	0.005	0.008
GPT-3.5	10%	Debiased	0.046	0.011	0.048	0.030	0.060
GPT-3.5	25%	LLM	0.002	0.001	0.002	0.001	0.003
GPT-3.5	25%	Human Validation	0.003	0.001	0.003	0.002	0.003
GPT-3.5	25%	Debiased	0.020	0.005	0.020	0.013	0.026
GPT-3.5	50%	LLM	0.002	0.001	0.002	0.001	0.003
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.001	0.002
GPT-3.5	50%	Debiased	0.014	0.003	0.015	0.009	0.019
GPT-4o	5%	LLM	0.001	0.000	0.001	0.001	0.002
GPT-4o	5%	Human Validation	0.014	0.003	0.015	0.010	0.017
GPT-4o	5%	Debiased	0.273	0.804	0.080	0.050	0.778
GPT-4o	10%	LLM	0.001	0.000	0.001	0.001	0.002
GPT-4o	10%	Human Validation	0.007	0.001	0.007	0.005	0.009
GPT-4o	10%	Debiased	0.029	0.008	0.031	0.019	0.039
GPT-4o	25%	LLM	0.001	0.000	0.001	0.001	0.002
GPT-4o	25%	Human Validation	0.003	0.001	0.003	0.002	0.003
GPT-4o	25%	Debiased	0.013	0.003	0.013	0.009	0.017
GPT-4o	50%	LLM	0.001	0.000	0.001	0.001	0.002
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.001	0.002
GPT-4o	50%	Debiased	0.010	0.002	0.010	0.007	0.013

Table 33: MSE Summary Statistics of β_t, t =Defense, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	5%	Human Validation	0.008	0.002	0.009	0.005	0.011
GPT-3.5	5%	Debiased	0.020	0.007	0.019	0.010	0.030
GPT-3.5	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
GPT-3.5	10%	Debiased	0.009	0.003	0.009	0.004	0.013
GPT-3.5	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	Human Validation	0.002	0.000	0.002	0.001	0.002
GPT-3.5	25%	Debiased	0.004	0.001	0.004	0.002	0.007
GPT-3.5	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Debiased	0.004	0.001	0.004	0.002	0.005
GPT-4o	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	5%	Human Validation	0.008	0.002	0.009	0.005	0.011
GPT-4o	5%	Debiased	0.016	0.005	0.018	0.009	0.023
GPT-4o	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
GPT-4o	10%	Debiased	0.008	0.002	0.008	0.004	0.011
GPT-4o	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	25%	Human Validation	0.002	0.000	0.002	0.001	0.002
GPT-4o	25%	Debiased	0.004	0.001	0.004	0.002	0.005
GPT-4o	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Debiased	0.003	0.001	0.004	0.002	0.005

Table 34: MSE Summary Statistics of β_t, t =Government Operations, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	5%	Human Validation	0.006	0.001	0.006	0.004	0.008
GPT-3.5	5%	Debiased	0.108	0.521	0.021	0.012	0.042
GPT-3.5	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	10%	Human Validation	0.003	0.001	0.003	0.002	0.004
GPT-3.5	10%	Debiased	0.009	0.003	0.009	0.006	0.013
GPT-3.5	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-3.5	25%	Debiased	0.004	0.001	0.005	0.003	0.006
GPT-3.5	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Debiased	0.004	0.001	0.004	0.002	0.005
GPT-4o	5%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	5%	Human Validation	0.006	0.001	0.006	0.004	0.008
GPT-4o	5%	Debiased	0.013	0.005	0.013	0.008	0.018
GPT-4o	10%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	10%	Human Validation	0.003	0.001	0.003	0.002	0.004
GPT-4o	10%	Debiased	0.006	0.001	0.006	0.004	0.007
GPT-4o	25%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	25%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-4o	25%	Debiased	0.003	0.001	0.003	0.002	0.004
GPT-4o	50%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Debiased	0.002	0.001	0.003	0.002	0.003

Table 35: MSE Summary Statistics of β_t, t =Public Lands and Water Management, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	5%	Human Validation	0.007	0.002	0.008	0.005	0.009
GPT-3.5	5%	Debiased	0.011	0.008	0.010	0.005	0.016
GPT-3.5	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
GPT-3.5	10%	Debiased	0.005	0.002	0.005	0.003	0.008
GPT-3.5	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	Human Validation	0.001	0.000	0.002	0.001	0.002
GPT-3.5	25%	Debiased	0.003	0.001	0.003	0.001	0.004
GPT-3.5	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Debiased	0.002	0.001	0.002	0.001	0.003
GPT-4o	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	5%	Human Validation	0.007	0.002	0.008	0.005	0.009
GPT-4o	5%	Debiased	0.007	0.002	0.008	0.004	0.010
GPT-4o	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
GPT-4o	10%	Debiased	0.004	0.001	0.004	0.002	0.005
GPT-4o	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	25%	Human Validation	0.001	0.000	0.002	0.001	0.002
GPT-4o	25%	Debiased	0.002	0.001	0.002	0.001	0.003
GPT-4o	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Debiased	0.002	0.001	0.002	0.001	0.002

Table 36: MSE Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%

Table 37: Coverage Summary Statistics of β_t, t =Health, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.867	0.096	0.911	0.689	0.952
5%	Human Validation	0.919	0.008	0.919	0.906	0.931
5%	Debiased	0.920	0.011	0.921	0.904	0.939
10%	LLM	0.866	0.097	0.913	0.679	0.947
10%	Human Validation	0.935	0.008	0.935	0.923	0.947
10%	Debiased	0.940	0.007	0.940	0.928	0.952
25%	LLM	0.865	0.097	0.909	0.692	0.948
25%	Human Validation	0.944	0.008	0.944	0.931	0.954
25%	Debiased	0.945	0.007	0.946	0.934	0.955
50%	LLM	0.865	0.099	0.913	0.653	0.947
50%	Human Validation	0.949	0.007	0.948	0.939	0.959
50%	Debiased	0.948	0.007	0.949	0.937	0.957

Table 38: Coverage Summary Statistics of β_t, t =Banking, Finance, and Domestic Commerce, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.685	0.225	0.764	0.252	0.932
5%	Human Validation	0.917	0.009	0.917	0.901	0.930
5%	Debiased	0.938	0.010	0.939	0.920	0.953
10%	LLM	0.689	0.226	0.770	0.242	0.937
10%	Human Validation	0.933	0.008	0.934	0.918	0.945
10%	Debiased	0.940	0.008	0.942	0.926	0.951
25%	LLM	0.686	0.225	0.759	0.237	0.938
25%	Human Validation	0.943	0.008	0.944	0.929	0.955
25%	Debiased	0.946	0.007	0.946	0.937	0.957
50%	LLM	0.690	0.225	0.766	0.255	0.935
50%	Human Validation	0.947	0.007	0.948	0.936	0.959
50%	Debiased	0.948	0.007	0.949	0.937	0.958

Table 39: Coverage Summary Statistics of β_t, t =Defense, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.925	0.032	0.936	0.879	0.957
5%	Human Validation	0.932	0.009	0.931	0.919	0.946
5%	Debiased	0.932	0.010	0.933	0.919	0.950
10%	LLM	0.926	0.030	0.938	0.876	0.953
10%	Human Validation	0.940	0.008	0.941	0.927	0.951
10%	Debiased	0.941	0.008	0.941	0.928	0.955
25%	LLM	0.925	0.033	0.938	0.868	0.953
25%	Human Validation	0.946	0.007	0.946	0.934	0.958
25%	Debiased	0.947	0.007	0.947	0.934	0.957
50%	LLM	0.926	0.031	0.933	0.883	0.956
50%	Human Validation	0.947	0.006	0.947	0.938	0.958
50%	Debiased	0.947	0.007	0.947	0.934	0.957

Table 40: Coverage Summary Statistics of β_t, t =Government Operations, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.891	0.065	0.915	0.740	0.954
5%	Human Validation	0.937	0.009	0.938	0.926	0.953
5%	Debiased	0.944	0.009	0.944	0.931	0.956
10%	LLM	0.890	0.066	0.912	0.750	0.949
10%	Human Validation	0.943	0.008	0.943	0.930	0.954
10%	Debiased	0.947	0.008	0.947	0.934	0.957
25%	LLM	0.894	0.064	0.915	0.737	0.954
25%	Human Validation	0.947	0.007	0.947	0.938	0.958
25%	Debiased	0.947	0.008	0.948	0.935	0.959
50%	LLM	0.889	0.066	0.913	0.735	0.947
50%	Human Validation	0.951	0.007	0.950	0.940	0.961
50%	Debiased	0.947	0.007	0.948	0.935	0.957

Table 41: Coverage Summary Statistics of β_t, t =Public Lands and Water Management, by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.901	0.062	0.923	0.784	0.956
5%	Human Validation	0.938	0.007	0.938	0.926	0.950
5%	Debiased	0.938	0.008	0.938	0.926	0.951
10%	LLM	0.902	0.056	0.920	0.785	0.953
10%	Human Validation	0.943	0.007	0.942	0.932	0.956
10%	Debiased	0.945	0.007	0.944	0.934	0.956
25%	LLM	0.903	0.058	0.925	0.783	0.952
25%	Human Validation	0.948	0.007	0.948	0.937	0.959
25%	Debiased	0.947	0.007	0.947	0.937	0.959
50%	LLM	0.904	0.055	0.923	0.796	0.954
50%	Human Validation	0.950	0.007	0.951	0.938	0.960
50%	Debiased	0.949	0.007	0.948	0.939	0.961

Table 42: Coverage Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%

Table 43: Coverage Summary Statistics of β_t, t =Health, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.812	0.108	0.804	0.642	0.938
GPT-3.5	5%	Human Validation	0.918	0.008	0.917	0.905	0.931
GPT-3.5	5%	Debiased	0.926	0.010	0.927	0.913	0.941
GPT-3.5	10%	LLM	0.811	0.110	0.810	0.636	0.934
GPT-3.5	10%	Human Validation	0.935	0.007	0.937	0.923	0.945
GPT-3.5	10%	Debiased	0.940	0.009	0.940	0.925	0.955
GPT-3.5	25%	LLM	0.810	0.109	0.800	0.623	0.936
GPT-3.5	25%	Human Validation	0.942	0.007	0.942	0.929	0.955
GPT-3.5	25%	Debiased	0.946	0.006	0.946	0.934	0.955
GPT-3.5	50%	LLM	0.809	0.113	0.820	0.621	0.937
GPT-3.5	50%	Human Validation	0.949	0.008	0.948	0.938	0.961
GPT-3.5	50%	Debiased	0.946	0.007	0.946	0.937	0.956
GPT-4o	5%	LLM	0.923	0.030	0.932	0.874	0.958
GPT-4o	5%	Human Validation	0.920	0.008	0.920	0.908	0.933
GPT-4o	5%	Debiased	0.914	0.009	0.913	0.901	0.928
GPT-4o	10%	LLM	0.922	0.027	0.933	0.871	0.950
GPT-4o	10%	Human Validation	0.935	0.008	0.935	0.923	0.948
GPT-4o	10%	Debiased	0.940	0.005	0.940	0.929	0.947
GPT-4o	25%	LLM	0.921	0.030	0.934	0.875	0.951
GPT-4o	25%	Human Validation	0.946	0.007	0.946	0.933	0.954
GPT-4o	25%	Debiased	0.945	0.007	0.946	0.933	0.954
GPT-4o	50%	LLM	0.921	0.027	0.929	0.874	0.948
GPT-4o	50%	Human Validation	0.948	0.007	0.948	0.939	0.958
GPT-4o	50%	Debiased	0.949	0.006	0.951	0.940	0.957

Table 44: Coverage Summary Statistics of β_t, t =Banking, Finance, and Domestic Commerce, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.596	0.255	0.650	0.178	0.903
GPT-3.5	5%	Human Validation	0.917	0.010	0.917	0.901	0.933
GPT-3.5	5%	Debiased	0.944	0.006	0.944	0.936	0.956
GPT-3.5	10%	LLM	0.599	0.255	0.667	0.200	0.913
GPT-3.5	10%	Human Validation	0.932	0.008	0.932	0.916	0.944
GPT-3.5	10%	Debiased	0.942	0.010	0.943	0.926	0.953
GPT-3.5	25%	LLM	0.599	0.256	0.659	0.172	0.915
GPT-3.5	25%	Human Validation	0.943	0.008	0.944	0.931	0.953
GPT-3.5	25%	Debiased	0.946	0.009	0.947	0.933	0.959
GPT-3.5	50%	LLM	0.602	0.255	0.649	0.184	0.921
GPT-3.5	50%	Human Validation	0.947	0.007	0.948	0.937	0.959
GPT-3.5	50%	Debiased	0.948	0.006	0.949	0.939	0.957
GPT-4o	5%	LLM	0.775	0.147	0.777	0.540	0.941
GPT-4o	5%	Human Validation	0.917	0.008	0.917	0.905	0.928
GPT-4o	5%	Debiased	0.933	0.010	0.935	0.920	0.946
GPT-4o	10%	LLM	0.779	0.150	0.792	0.524	0.950
GPT-4o	10%	Human Validation	0.934	0.008	0.935	0.919	0.945
GPT-4o	10%	Debiased	0.939	0.007	0.940	0.928	0.949
GPT-4o	25%	LLM	0.774	0.146	0.786	0.536	0.940
GPT-4o	25%	Human Validation	0.943	0.008	0.944	0.928	0.955
GPT-4o	25%	Debiased	0.946	0.006	0.944	0.938	0.957
GPT-4o	50%	LLM	0.777	0.148	0.793	0.537	0.939
GPT-4o	50%	Human Validation	0.947	0.008	0.948	0.934	0.958
GPT-4o	50%	Debiased	0.948	0.008	0.949	0.936	0.959

Table 45: Coverage Summary Statistics of β_t, t =Defense, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.914	0.038	0.923	0.826	0.954
GPT-3.5	5%	Human Validation	0.932	0.008	0.932	0.923	0.944
GPT-3.5	5%	Debiased	0.938	0.008	0.938	0.927	0.951
GPT-3.5	10%	LLM	0.915	0.036	0.926	0.853	0.950
GPT-3.5	10%	Human Validation	0.940	0.008	0.942	0.926	0.953
GPT-3.5	10%	Debiased	0.943	0.008	0.944	0.930	0.956
GPT-3.5	25%	LLM	0.913	0.039	0.923	0.832	0.950
GPT-3.5	25%	Human Validation	0.946	0.007	0.946	0.934	0.959
GPT-3.5	25%	Debiased	0.947	0.005	0.947	0.939	0.955
GPT-3.5	50%	LLM	0.915	0.037	0.921	0.836	0.954
GPT-3.5	50%	Human Validation	0.947	0.005	0.945	0.939	0.954
GPT-3.5	50%	Debiased	0.948	0.006	0.947	0.936	0.958
GPT-4o	5%	LLM	0.936	0.019	0.940	0.892	0.957
GPT-4o	5%	Human Validation	0.932	0.010	0.931	0.916	0.947
GPT-4o	5%	Debiased	0.927	0.009	0.926	0.912	0.941
GPT-4o	10%	LLM	0.937	0.017	0.943	0.903	0.953
GPT-4o	10%	Human Validation	0.940	0.007	0.940	0.928	0.948
GPT-4o	10%	Debiased	0.939	0.009	0.940	0.925	0.952
GPT-4o	25%	LLM	0.937	0.019	0.942	0.901	0.958
GPT-4o	25%	Human Validation	0.947	0.008	0.947	0.934	0.956
GPT-4o	25%	Debiased	0.946	0.008	0.946	0.933	0.958
GPT-4o	50%	LLM	0.937	0.019	0.943	0.895	0.956
GPT-4o	50%	Human Validation	0.947	0.006	0.948	0.938	0.959
GPT-4o	50%	Debiased	0.945	0.007	0.948	0.933	0.955

Table 46: Coverage Summary Statistics of β_t, t =Government Operations, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.865	0.078	0.899	0.715	0.948
GPT-3.5	5%	Human Validation	0.937	0.007	0.938	0.928	0.947
GPT-3.5	5%	Debiased	0.945	0.008	0.946	0.932	0.956
GPT-3.5	10%	LLM	0.866	0.078	0.897	0.719	0.944
GPT-3.5	10%	Human Validation	0.943	0.009	0.943	0.930	0.957
GPT-3.5	10%	Debiased	0.945	0.009	0.946	0.934	0.957
GPT-3.5	25%	LLM	0.868	0.078	0.893	0.715	0.949
GPT-3.5	25%	Human Validation	0.948	0.007	0.948	0.938	0.958
GPT-3.5	25%	Debiased	0.948	0.008	0.948	0.936	0.960
GPT-3.5	50%	LLM	0.864	0.081	0.895	0.717	0.943
GPT-3.5	50%	Human Validation	0.951	0.007	0.950	0.940	0.963
GPT-3.5	50%	Debiased	0.946	0.008	0.946	0.934	0.958
GPT-4o	5%	LLM	0.917	0.034	0.927	0.853	0.957
GPT-4o	5%	Human Validation	0.938	0.010	0.939	0.917	0.953
GPT-4o	5%	Debiased	0.942	0.009	0.941	0.928	0.956
GPT-4o	10%	LLM	0.914	0.038	0.928	0.848	0.950
GPT-4o	10%	Human Validation	0.943	0.007	0.942	0.932	0.952
GPT-4o	10%	Debiased	0.948	0.008	0.948	0.935	0.958
GPT-4o	25%	LLM	0.919	0.030	0.929	0.862	0.956
GPT-4o	25%	Human Validation	0.946	0.007	0.945	0.936	0.956
GPT-4o	25%	Debiased	0.947	0.008	0.947	0.935	0.958
GPT-4o	50%	LLM	0.914	0.031	0.921	0.860	0.950
GPT-4o	50%	Human Validation	0.951	0.006	0.950	0.941	0.960
GPT-4o	50%	Debiased	0.948	0.005	0.948	0.938	0.954

Table 47: Coverage Summary Statistics of β_t, t =Public Lands and Water Management, by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.913	0.053	0.930	0.822	0.956
GPT-3.5	5%	Human Validation	0.938	0.006	0.938	0.930	0.946
GPT-3.5	5%	Debiased	0.941	0.007	0.942	0.929	0.953
GPT-3.5	10%	LLM	0.911	0.049	0.927	0.813	0.953
GPT-3.5	10%	Human Validation	0.942	0.008	0.941	0.932	0.956
GPT-3.5	10%	Debiased	0.944	0.008	0.941	0.934	0.958
GPT-3.5	25%	LLM	0.912	0.051	0.930	0.821	0.952
GPT-3.5	25%	Human Validation	0.946	0.007	0.946	0.937	0.956
GPT-3.5	25%	Debiased	0.948	0.006	0.948	0.938	0.960
GPT-3.5	50%	LLM	0.912	0.050	0.929	0.818	0.955
GPT-3.5	50%	Human Validation	0.950	0.006	0.951	0.941	0.959
GPT-3.5	50%	Debiased	0.948	0.008	0.946	0.937	0.963
GPT-4o	5%	LLM	0.889	0.068	0.902	0.796	0.952
GPT-4o	5%	Human Validation	0.937	0.009	0.937	0.924	0.950
GPT-4o	5%	Debiased	0.936	0.008	0.936	0.924	0.947
GPT-4o	10%	LLM	0.893	0.062	0.915	0.794	0.953
GPT-4o	10%	Human Validation	0.944	0.006	0.943	0.935	0.955
GPT-4o	10%	Debiased	0.945	0.006	0.946	0.934	0.955
GPT-4o	25%	LLM	0.893	0.063	0.914	0.792	0.951
GPT-4o	25%	Human Validation	0.949	0.007	0.950	0.937	0.962
GPT-4o	25%	Debiased	0.946	0.007	0.945	0.936	0.956
GPT-4o	50%	LLM	0.895	0.060	0.917	0.796	0.951
GPT-4o	50%	Human Validation	0.950	0.008	0.950	0.936	0.963
GPT-4o	50%	Debiased	0.949	0.007	0.949	0.940	0.959

Table 48: Coverage Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = \sum_t Y_t \beta_t$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%